



**THE YENEPOYA INSTITUTE OF ARTS, SCIENCE,
COMMERCE AND MANAGEMENT
A CONSTITUENT UNIT OF YENEPOYA (DEEMED TO BE UNIVERSITY)**

**AN INTERNSHIP REPORT ON
DELETED FILE RECOVERY**

SUBMITTED BY

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VI SEMESTER

**BCA (CYBER SECURITY, ETHICAL HACKING AND DIGITAL
FORENSICS)**

UNDER THE GUIDANCE OF

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TECHNOLOGY**

**IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR
THE AWARD**

OF

THE DEGREE OF BACHELOR IN COMPUTER APPLICATIONS

2026



CERTIFICATE

This is to certify that Ms. **DEVIKA DAS**, bearing Registration Number **23BBCCED045**, of Batch **2023–2026**, is a bonafide student of the Bachelor of Computer Applications (BCA) programme at Yenepoya Institute of Arts, Science, Commerce and Management, affiliated to Yenepoya (Deemed to be University).

He/She has successfully completed the internship titled “**DELETED FILE RECOVERY**” at **IBM, BANGALORE** under the guidance of **Mrs. E. ALEKHYA**.

This internship report is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications at Yenepoya (Deemed to be University), Bengaluru, Karnataka.

Signature of Internal Guide

Signature of HOD

Signature of Vice Principal

Signature of Director-Academics



DECLARATION

I, Mr./Ms. **DEVIKA DAS**, bearing Registration Number **23BBCCED045**, hereby declare that the internship report titled “**DELETED FILE RECOVERY**”, with reference to **IBM**, has been prepared by me in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) at Yenepoya (Deemed to be University).

I further declare that this internship report is the result of my own work carried out under the guidance of **Mrs E. Alekhya, Assistant Professor**, Yenepoya Institute of Arts, Science, Commerce and Management.

I also confirm that this report has not been previously submitted to any university or institution for the award of any degree or diploma.

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ABSTRACT

In today's digital world, data has become one of the most valuable assets for any individual, organization, and government. In recent times, due to increased reliance on digital storage mediums and cloud computing, data loss scenarios like unintentional deletion, insider threats, cyber-attacks, ransomware, and system crashes have become very common. Contrary to popular belief, deleting files from digital storage mediums does not necessarily mean that they are lost forever. In fact, in all operating systems, deleted files can be recovered unless they are overwritten. This has become a major security concern, and at the same time, it has also become an opportunity in digital forensics.

The topic for the internship project is "Deleted File Recovery and Security Analysis Using Autopsy." This is a project based on the recovery and security analysis of deleted files using Autopsy, a digital forensic tool. This is an internship project carried out in a controlled environment, simulating scenarios related to file deletion and digital forensic analysis. This study is related to the internal working of file systems like NTFS, data structures like MFT, and the way deleted files are stored in the unallocated space. This project is based on interdisciplinary ideas related to Digital Forensics, Cyber Security, Ethical Hacking, and Cloud Computing. From a digital forensic point of view, the study is related to the recovery of deleted files with related information like metadata, date, and file attributes. From a cybersecurity point of view, the study is related to the security implications of file deletion and the risk associated with data exposure. From an ethical hacking point of view, the study is related to the importance of data sanitization and the risk associated with the exposure of data. Remanence issues in cloud computing are also discussed in the study. With hands-on practice, deleted files of various types have been successfully recovered using Autopsy's ingest modules and reporting tools. The research assesses the recovery efficiency of deleted files, various factors influencing recoverability, and limitations of recovery processes including the use of SSDs with TRIM support and encryption. The findings of the project clearly emphasize the need to implement proper data destruction techniques, encryption techniques, and storage management in recovery not only strengthens digital forensic investigation capabilities but also enhances organizational cybersecurity strategies and cloud data protection frameworks.

CHAPTER 1:

INTRODUCTION OF DIGITAL FORENSICS

The exponential development of digital technologies has transformed the way in which people, organizations, and governments operate. The present society has come to rely on computers, mobile technologies, cloud computing, and network technologies for communication, businesses, governance, and information handling. As the digital revolution accelerates, tremendous volumes of sensitive and mission-critical information are constantly being created, stored, and transmitted using the interconnected technologies.

The development of digital technologies has been characterized by a tremendous rise in the number of cyber-enabled crimes and digital security incidents. Cyber criminals exploit the vulnerabilities of the technologies to perpetuate crimes such as unauthorized access, financial crimes, intellectual property crimes, ransomware, and data breach. Digital crimes are distinct from conventional crimes in that digital crimes leave intangible evidence that requires special knowledge to understand.

However, digital forensics has emerged to be a critical tool in addressing the concerns. It is a science of gathering, analysing, and preserving digital evidence in a way that ensures its integrity and admissibility in judicial proceedings. Digital forensics incorporates computer science, cyber security, and law to ensure the development of tools for digital reconstruction, recovery of deleted data, and judicial support of scientifically informed evidence.

This chapter presents an overview of digital forensics, which encompasses its history, need in the modern era, contribution to cybercrime, various forms of digital evidence, operational problems, and legal and ethical issues of digital forensic practice.

1.1 EVOLUTION OF DIGITAL FORENSICS

Digital forensic science was first recognized in the late 1980s due to the increased use of computers in business, government, and personal activities. The first phase of digital investigation involved simple data recovery methods for individual computer systems. Computer crimes in this era of digital forensic science ranged from unauthorized access, data tampering, and fraud. Digital forensic science in this era involved digital media investigation, which included examining digital media such as floppy disks and hard disks for deleted data and suspicious data.

The rapid development of the internet in the 1990s marked another milestone in digital crime. The activities of cybercrime have been extensive and sophisticated. Computer crimes, which involved hacking, identity theft, internet fraud, theft of intellectual property, and viruses, have started to increase dramatically. The conventional investigation methods were not enough to deal with digital crimes. Thus, there was a need to develop a set of standard digital forensic techniques.

- Cloud Forensics – Investigation of data stored in distributed and virtualized cloud computing environments.
- Database Forensics – Analysis of structured data systems and transaction logs.
- Malware and Memory Analysis – Examination of malicious code and memory artifacts.

The rising complexity of digital ecosystems like IoT devices, blockchain technologies, and artificial intelligence technologies has further widened the scope of digital forensic investigations.

Significant advancements in artificial intelligence technologies, machine learning technologies, big data analytics, and forensic automation technologies have further strengthened digital forensic investigations.

Presently, digital forensic science is an integral part of modern criminal justice systems, cybercrime units, national security organizations, and corporate incident response teams. It plays a vital role not only in preventing crimes but also in preventing cyber security breaches, developing stronger compliance models, and supporting litigation processes.

1.2 NEED FOR DIGITAL INVESTIGATION

The rapid digitalization of day-to-day activities has revolutionized the face of investigations. Today, communication, financial transactions, and business and social interactions are mainly done electronically. This has, therefore, elevated the role of digital evidence in the administration of justice in both criminal and civil cases. Virtually, any case, be it one of fraud, harassment, intellectual property, financial impropriety, or cybercrime, has some element of digital evidence. This digital evidence may be in the form of emails, transaction records, security cameras, or online communication records, among others.

Unlike traditional forms of evidence, digital evidence has been shown to be volatile and easily susceptible to alteration. This has necessitated the development of specific methodologies for the collection, analysis, and presentation of digital evidence in the administration of justice.

The need for digital investigations has been necessitated by the following critical factors:

Increase in Frequency and Sophistication of Cybercrimes: Today, cybercrimes are being committed in a sophisticated manner using techniques such as encryption, anonymization networks, ransomware attacks, and multi-stage attacks. Simple methods of investigation are not enough to investigate technologically sophisticated cybercrimes.

Popularization of Encrypted Communication Tools: Messaging tools and email applications with end-to-end encryption and virtual private networks (VPNs) are being widely used to increase user privacy.

Expansion of Cloud Computing and Remote Data Storage: It is not necessary for data to be stored locally; it is stored remotely in cloud environments in various locations. For data to be recovered efficiently, investigators must be aware of cloud computing and data synchronization techniques.

Growth of Digital Financial Systems and E-Commerce: Digital financial transactions and e-commerce have given rise to various types of financial cybercrimes and cyber-

enabled offenses. Digital investigation is required to follow various types of financial transactions.

Judicial Requirement for Validated Digital Evidence: Digital evidence is used in various legal cases and trials. Digital investigation is required to ensure that data is collected and processed in a scientific manner and is admissible in court.

Digital investigation is vital in reconstructing the sequence of events in crime occurrences, determining liable parties, and establishing data origins and circulation. By examining digital data and establishing correlations between events and actors in digital environments, investigators can establish links between actions and actors in digital environments.

It is apparent that without digital investigation methodologies, modern investigations would not be technically sound in establishing accountability in technology-driven crimes. As digital environments increasingly permeate all aspects of our lives, digital investigation is not only desirable but essential in establishing justice and corporate and cyber security resilience.

1.3 ROLE IN CYBER CRIME INVESTIGATION

Digital forensics has an essential and indispensable role to play in the investigation of cybercrimes through its methodologies of identification, preservation, examination, and analysis of digital evidence. With cybercrimes targeting digital infrastructures, almost every cybercrime has a digital component to it. Digital forensic investigators undertake a range of technical and procedural activities, including:

Acquisition of Forensic Images: Making bit-for-bit replicas of digital storage media using various forensic image acquisition techniques. This is to ensure that examination is done on a duplicate copy without compromising the original data.

Recovery of Deleted, Hidden, or Encrypted Files: Using various data recovery techniques to recover data from unallocated space or partially overwritten data on digital media. Investigators may also try to recover password information through legal means.

Metadata and Timestamp Analysis: Investigation of timestamp data to reconstruct event sequences. Analysis of metadata helps to identify user behavior patterns and anomalies.

Network Traffic Examination: Investigation of network traffic to identify suspicious activities or unauthorized access to computer systems or networks through various firewall logs and intrusion detection systems.

Tracing IP Addresses and Digital Footprints: Investigation to identify the source or path of cyber activities through IP address tracking and service provider logs.
Reconstruction of User

Activities and System Events: Rebuilding sequences of actions performed on a system, including login attempts, file transfers, application usage, and registry modifications.

A fundamental principle in digital forensics is the preservation of evidence integrity. During evidence acquisition, investigators employ **write blockers** to prevent any modification to the original storage media. Additionally, cryptographic hash functions such as SHA-256 are used to generate unique digital fingerprints of evidence files. Any

alteration in the data would result in a change in the hash value, thereby ensuring authenticity and reliability. The maintenance of a strict chain of custody is equally important. Every action taken with regard to the evidence is properly documented, and the date, time, people, and equipment used are all taken into account. This ensures that transparency is maintained, and the evidence is admissible in a court of law.

Digital forensics acts as a bridge that connects technological investigation with legal adjudication by combining technology and law. It takes complex information from technology and presents it in a structured and comprehensible manner, which is easily admissible in a court of law. Thus, it is not just useful for identifying criminals, but it is equally useful for upholding the principles of justice in a technology-dominated world.

1.4 TYPES OF DIGITAL EVIDENCE

Digital evidence is defined as any information of probative value that is in a digital form and is used in the investigation. Unlike any other evidence, digital evidence is in a binary form and may be scattered over several devices, networks, and cloud environments. The nature of digital evidence is such that it is intangible, and the complexity of dealing with it makes it both a challenge and a concern from a legal perspective.

Digital evidence is found in a variety of sources, from physical storage devices such as hard drives, SSDs, and USB devices, mobile devices, network servers, virtual machines, cloud environments, and even Internet of Things devices. The nature of digital evidence is such that it is created by devices and systems that operate and function, and even the most insignificant information may be of significant importance during the investigation.

Major Categories of Digital Evidence:

Digital evidence can be broadly classified into the following categories:

1. Files and Documents

This includes structured and unstructured files such as PDFs, word processing documents, spreadsheets, presentations, and executables. These documents may carry embedded metadata indicating authorship, creation dates, modification history, and system details..

2. Multimedia Content

Images, videos, and audio files frequently give a direct visual or auditory evidence of activities. Multimedia files are also found to have metadata that can show timestamps, device identifiers, as well as geolocation information

3. Emails and Messaging Records

Electronic communication records, including email messages, chat logs, instant messaging records, and social media interactions, are often used as important evidence in fraud cases, harassment cases, and conspiracy cases. Header analysis of email messages may reveal routing paths and IP addresses.. System Logs and Event Logs

Operating systems and applications produce log data that tracks user activities, login attempts, system failures, configuration changes, and network activities. Log analysis is a fundamental part of creating a timeline of activities and identifying suspicious activities. Browser History and Cookies Web browsing artifacts, cached files, cookies, and download histories provide insight into user behaviour, accessed websites, and potential interactions with malicious content.

4. Database Records

Databases store structured information such as transaction histories, customer records, and access logs. Forensic examination of databases may uncover unauthorized modifications, fraudulent transactions, or hidden entries.

5. GPS and Location Data

Modern devices and applications record geolocation data through GPS, Wi-Fi positioning, and cellular networks. Location data can be used to verify an individual's presence at a specific place and time.

6. Metadata

Metadata refers to “data about data,” including timestamps, file ownership details, access permissions, and modification history. Metadata often plays a crucial role in validating authenticity and detecting tampering.

7. Cloud Storage and Backup Data

With the widespread adoption of cloud services, digital evidence may reside in remote servers maintained by third-party providers. Cloud forensics requires understanding synchronization logs, access tokens, and distributed storage architectures.

Characteristics and Handling of Digital Evidence

Digital evidence possesses unique characteristics that differentiate it from physical evidence:

- **Volatility:** Some data, such as RAM contents and network traffic, is temporary and may be lost if not captured promptly.
- **Fragility:** Digital data can be altered, overwritten, or corrupted with minimal effort.
- **Replicability:** Exact copies can be created without degrading quality, which is both advantageous and challenging from a legal standpoint.
- **Susceptibility to Anti-Forensic Techniques:** Attackers may use encryption, log manipulation, or secure wiping tools to obscure traces.

Due to these characteristics, investigators must follow strict forensic acquisition procedures. This includes creating forensic images using write blockers, generating cryptographic hash values to verify integrity, maintaining detailed chain-of-custody documentation, and ensuring compliance with legal standards.

Proper handling of digital evidence ensures that it remains authentic, reliable, and admissible in judicial proceedings. Failure to adhere to standardized forensic protocols may result in evidence being challenged or rejected in court.

1.5 CHALLENGES IN DIGITAL FORENSICS

Yet, despite the advancements in technology and the development of sophisticated forensic tools, digital forensic science faces a plethora of operational, technical, and legal issues. The dynamic nature of cybercrimes and the rapid evolution of technology demand a dynamic response from investigators and law enforcement agencies.

1. Data Volume Explosion

Digital devices of today have enormous storage capacities, with some devices capable of storing terabytes of data. Additionally, organizations are capable of generating large volumes of data on a daily basis through various means, including logs, transactions, and interactions with cloud services. The exponential growth of digital evidence poses a challenge in terms of time required for acquisition and analysis of evidence.

2. Encryption Technologies

The use of sophisticated encryption techniques, including end-to-end encryption and full-disk encryption, provides an additional layer of security and enhances user privacy. However, in the absence of proper decryption mechanisms or authorization tools, investigators face a challenge in accessing encrypted evidence. extremely difficult. Encrypted messaging platforms, secure cloud storage, and password-protected devices often delay or limit forensic investigations.

3. Anti-Forensic Techniques

Cybercriminals are increasingly using anti-forensic techniques to avoid detection. These techniques include:

- Secure wiping of data and shredding of files
- Steganography (concealing data in images and media files)
- Tampering with log files and timestamps
- Anonymizing tools such as VPNs and proxy networks

It is a complex process to retrieve the data and requires expertise to spot any signs of manipulation.

4. Cloud and Distributed Systems

Cloud computing and distributed architectures store data across multiple servers and geographic regions. Evidence may be fragmented across different data centers, service providers, or jurisdictions. This decentralization complicates evidence collection, especially when service providers are located in foreign countries. Access often depends on mutual legal assistance treaties (MLATs) and international cooperation.

5. Rapid Technological Evolution

The development of new hardware devices, new versions of operating systems, new applications, and new versions of communication protocols necessitate constant enhancement of skills by those engaged in computer forensics.

6. Legal and Jurisdictional Complexities

Cybercrimes often involve a perpetrator from one country targeting victims from another country while using servers from a third country. This poses a problem from a legal point of view. How can evidence be seized? What are the data privacy laws? What is the jurisdiction? The harmonization of international cyber laws is a must to get over such hurdles.

1.6 LEGAL AND ETHICAL CONSIDERATIONS

Digital forensic investigations operate at the intersection of technology and law. Because such investigations involve access to sensitive personal, corporate, or governmental data, strict adherence to legal and ethical standards is mandatory. Any unauthorized access, procedural violation, or improper evidence handling may render the evidence inadmissible in court and expose investigators to legal liability.

The credibility of digital forensic findings depends not only on technical accuracy but also on procedural compliance and ethical integrity. Therefore, investigators must ensure that every phase of the forensic process from evidence identification to courtroom presentation aligns with established legal frameworks and professional codes of conduct.

1. Lawful Authorization

Before accessing digital systems, storage devices, or communication records, investigators must obtain proper legal authorization. This may include search warrants, court orders, or informed consent from the concerned parties. Unauthorized access to digital data constitutes a violation of privacy laws and may compromise the admissibility of collected evidence.

2. Chain of Custody Maintenance

The chain of custody refers to the documented record of how evidence is collected, transferred, stored, and analyzed. Every individual who handles the evidence must be identified, along with the date, time, and purpose of access. Proper documentation ensures transparency and prevents allegations of tampering or contamination.

3. Preservation of Data Integrity

Maintaining data integrity is a fundamental forensic principle. Investigators must ensure that digital evidence remains unchanged from the time of acquisition to presentation in court. This is typically achieved through:

- Use of write-blocking devices during evidence acquisition

- Creation of forensic images (bit-by-bit copies)
- Application of cryptographic hash functions (e.g., SHA-256) to generate unique digital fingerprints.

4. Protection of Privacy and Confidential Information

Digital investigations often expose personal communications, financial records, medical data, and confidential corporate information. Investigators must handle such data with strict confidentiality and limit access strictly to case-related purposes. Over-collection or misuse of data may violate privacy rights and data protection regulations.

5. Impartiality and Professional Integrity

Digital forensic experts must maintain neutrality and objectivity. Their role is to analyze and present facts, not to advocate for prosecution or defense. Biased analysis, selective reporting, or manipulation of findings undermines justice and professional credibility.

Investigators must base their conclusions solely on verifiable technical evidence and maintain transparency in reporting methodologies and limitations.

CHAPTER 2:

FILE SYSTEMS AND INTERNAL STRUCTURE

2.1 NTFS ARCHITECTURE

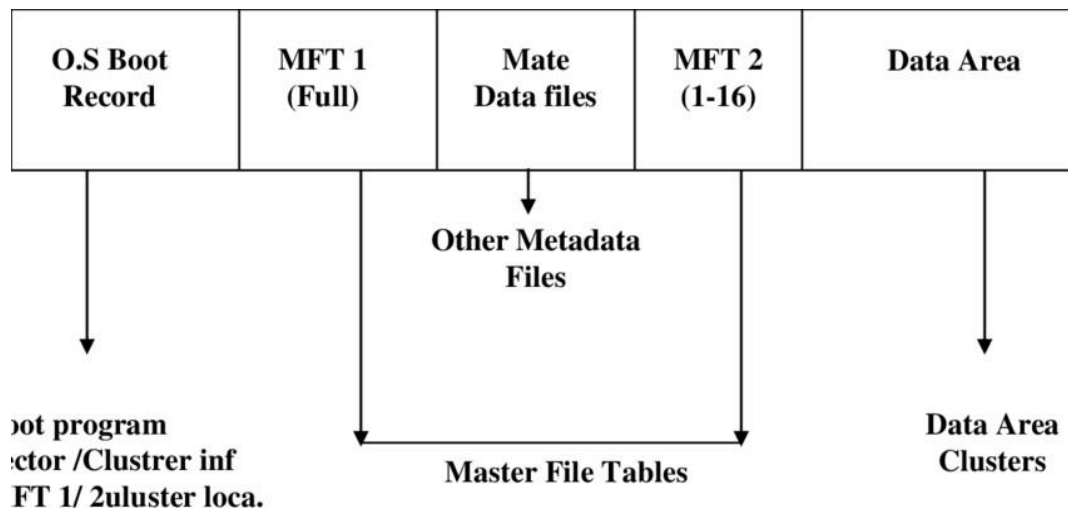


Figure 1: NTFS

The New Technology File System, abbreviated as NTFS, is the default file system used in the latest versions of Microsoft Windows operating systems. NTFS was first introduced in Windows NT and was developed to overcome the limitations of previous file systems, including improved reliability, security, scalability, and performance.

The NTFS file system is based on a structured file system architecture with a Master File Table (MFT) containing information about all files and directories on the volume. Unlike previous file systems, NTFS treats its metadata as files, enabling flexible and extensible data management. Each file is represented by a file record in the MFT, containing file attributes such as file name, date, and time stamps, and file data locations.

The main features of NTFS are:-

- Support for large file sizes and volumes
- File-level security through Access Control Lists (ACLs)
- Journaling capability for crash recovery
- Disk quotas and file compression
- Encryption using Encrypting File System (EFS)

NTFS stores data in clusters and uses advanced indexing techniques (such as B-trees) to optimize directory searches. It also maintains redundant metadata copies to improve fault tolerance.

From a digital forensic perspective, NTFS is highly valuable because it preserves detailed metadata, including file creation, modification, and access timestamps. Even deleted files may leave residual entries within the MFT, enabling partial recovery.

2.2 METADATA AND TIMESTAMPS

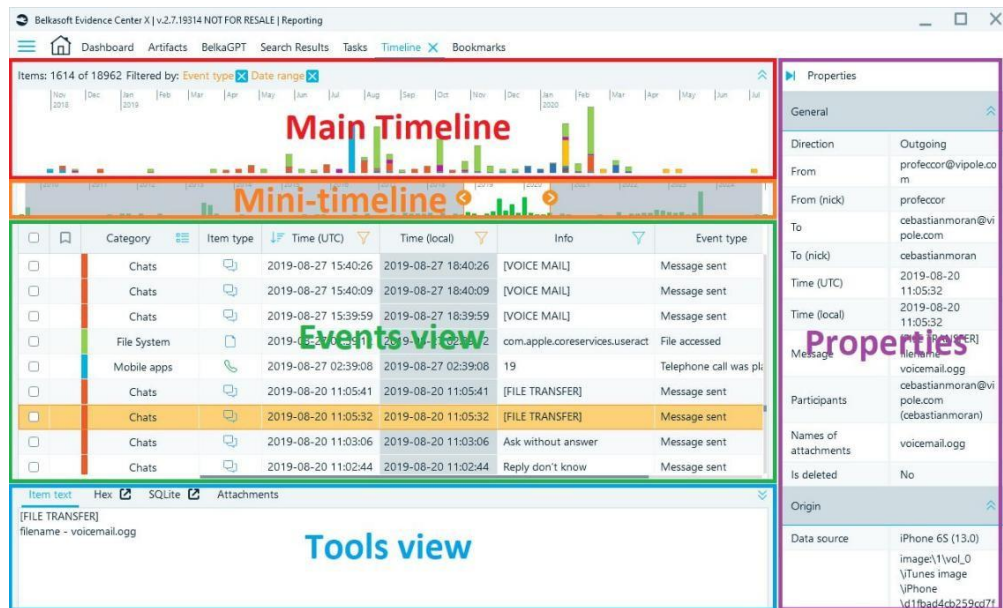


Figure 2: Metadata And Timestamps

Metadata refers to “data about data.” It provides descriptive information about files, including attributes and timestamps. In file systems, metadata is automatically generated and updated by the operating system.

The most significant timestamps in NTFS are often referred to as **MAC times**:

- **M – Modified Time:** When file content was last changed
- **A – Accessed Time:** When file was last opened or accessed
- **C – Changed Time:** When file metadata was last altered
- **B – Birth Time:** File creation time

Metadata is crucial in forensic timeline reconstruction. By analyzing timestamps, investigators can determine sequences of events, user activities, and possible tampering.

However, timestamps can be manipulated using specialized tools. Therefore, investigators cross-verify timestamp data with logs, registry entries, and other artifacts to ensure accuracy.

Metadata extends beyond file systems and includes EXIF data in images, document properties in PDFs, and system log details. This information often plays a decisive role in digital investigations.

2.3 JOURNALING MECHANISMS

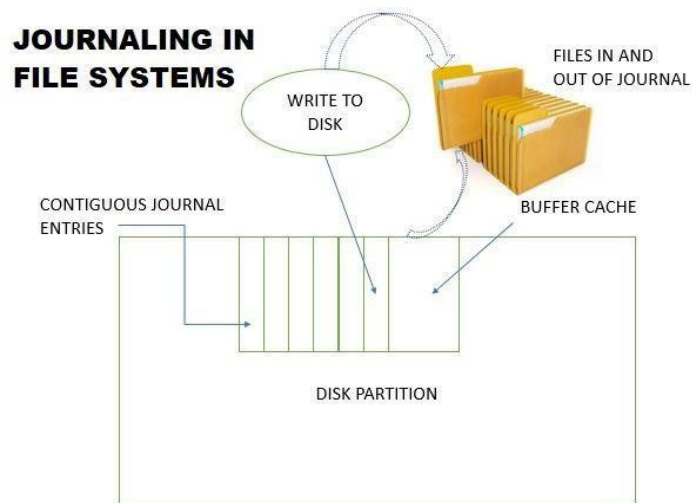


Figure 3: Journaling In File System

Journaling is a mechanism used by advanced file systems to maintain consistency and recover quickly after system crashes or power failures. Instead of directly writing changes to disk, the system first records intended operations in a journal (log file).

In NTFS, the journal file (\$Log File) records metadata changes before they are committed. If a crash occurs, the system replays the journal to restore consistency.

Benefits of journaling include:

- Reduced risk of file system corruption
- Faster recovery after crashes
- Improved data integrity

From a forensic perspective, journal files may contain traces of file creation, deletion, and modification activities. Even if a file has been deleted, journal entries may provide evidence of its previous existence.

2.4 FILE ALLOCATION AND CLUSTERING

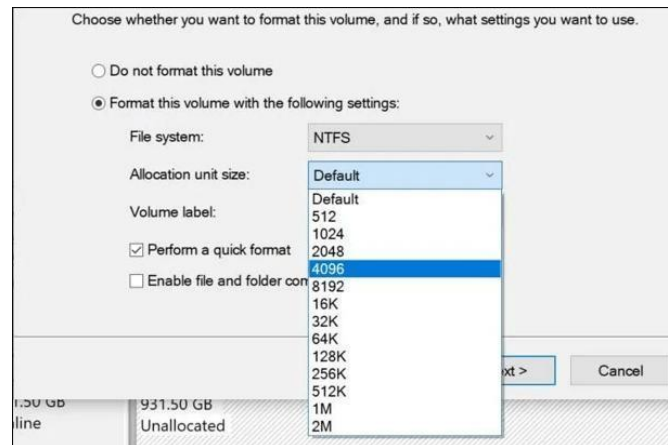


Figure 4: File Allocation And Clustering

File allocation refers to the method by which a file system stores and manages data blocks on disk. Storage media are divided into sectors, which are grouped into clusters (also known as allocation units).

A cluster is the smallest storage unit that can be allocated to a file. If a file is smaller than one cluster, the remaining space is unused but reserved for that file, leading to internal fragmentation.

File systems use allocation strategies such as:

- **Contiguous Allocation:** Stores file data in consecutive clusters.
- **Linked Allocation:** Uses pointers to link scattered clusters.
- **Indexed Allocation:** Maintains an index of cluster addresses.

In NTFS, file allocation information is stored within MFT attributes. Fragmentation occurs when file clusters are scattered across the disk, potentially reducing performance.

For forensic investigators, unallocated clusters and slack space are significant because they may contain remnants of deleted files. Data remnants stored in slack space can sometimes be recovered and used as evidence.

CHAPTER 3:

CONCEPTS OF FILE DELETION

3.1 LOGICAL VS PHYSICAL DELETION

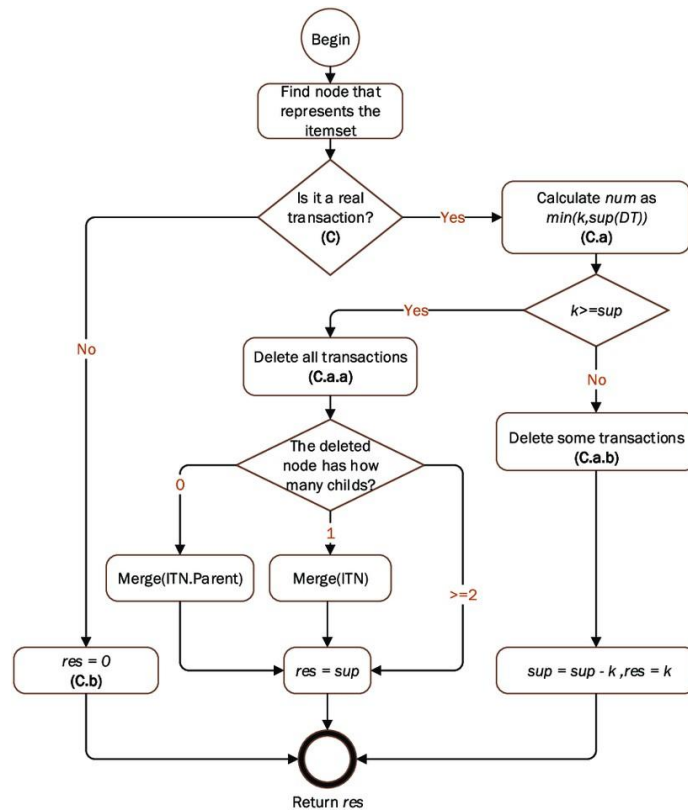


Figure 5: Flowchart

The deletion of files in today's operating systems does not directly delete the data on the storage media. Instead, it follows a two-step process: logical deletion and physical deletion.

Logical Deletion

Logical deletion takes place when an individual deletes a file, either by hitting the Delete button on their keyboard or emptying the Recycle Bin. At this point, the operating system removes the file from the directory and marks the clusters associated with that file as available. However, the actual file content remains intact on the disk until it is overwritten by new data. This is why recovery tools can restore deleted files if the storage space has not yet been reused.

Physical Deletion

Physical deletion occurs when the data blocks containing the file are overwritten or securely wiped. At this stage, the original binary content is replaced, making recovery extremely difficult or impossible.

Table 1: Logical vs Physical Deletion

FEATURE	LOGICAL DELETION	PHYSICAL DELETION
File Reference Removed	Yes	Yes
Actual Data Erased	No	Yes
Recovery Possible	High	Very Low
Time Required	Instant	Depends On Overwriting
Forensic Value	High	Minimal

provides a significant opportunity for recovery. Understanding this distinction is fundamental in digital investigations.

3.2 DATA REMANENCE



Figure 6: Inside Of Hard Disk Drive

Data remanence refers to the residual representation of digital data that remains on storage media even after attempts have been made to erase it. This phenomenon is particularly relevant in magnetic storage devices such as HDDs.

Even after logical deletion, magnetic traces of previously stored data may persist. Advanced forensic tools can sometimes recover fragments of these remnants.

Causes of Data Remanence

- Incomplete deletion
- Improper formatting
- Residual magnetic traces
- Partial overwriting

Table 2: Storage Type vs Remanence Risk

STORAGE TYPE	REMANENCE RISK	RECOVERY POSSIBILITY
HDD	High	Moderate To High
SSD (With Trim)	Low	Low
USB Flash	Moderate	Moderate

3.3 UNALLOCATED SPACE

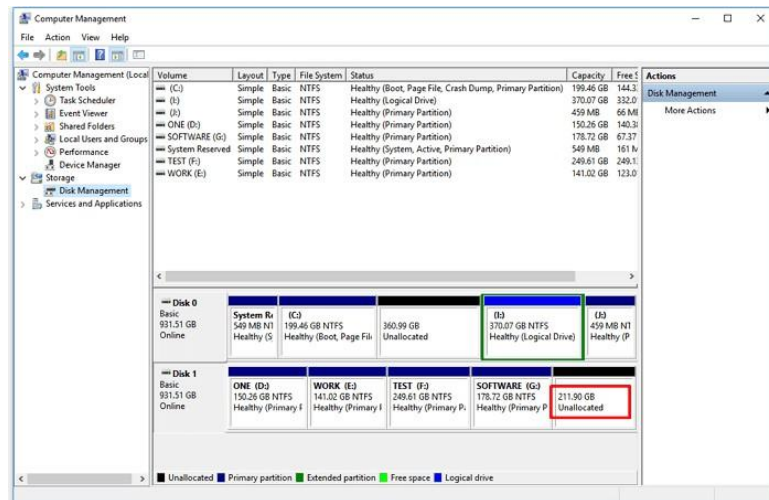


Figure 7: Un Allocated Space

Unallocated space refers to areas of a storage device not currently assigned to active files. When files are deleted logically, their clusters become part of unallocated space.

These areas may contain remnants of previously stored files, making them valuable in forensic investigations.

Characteristics of Unallocated Space

- Not visible to users
- Contains deleted file fragments
- Target area for forensic carving

Table 3: Allocated vs Unallocated Space

FEATURE	ALLOCATED SPACE	UNALLOCATED SPACE
Linked To Active Files	Yes	No
User Accessible	Yes	No
Contains Deleted Data	Rarely	Often
Forensic Importance	Moderate	Very High

3.4 SLACK SPACE

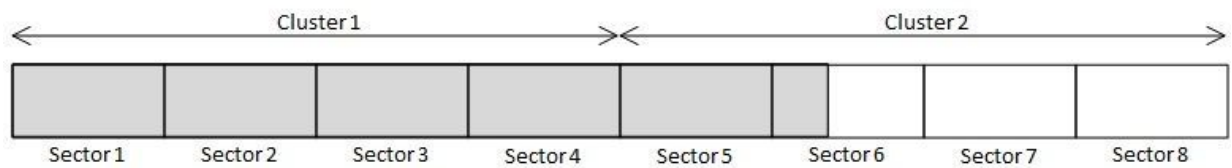


Figure 8: Slack Space

Slack space refers to unused space within allocated clusters. Since storage devices allocate space in clusters, if a file does not fully occupy its final cluster, the remaining portion becomes slack space.

Slack space may contain fragments of previously deleted files.

Types of Slack Space

- **RAM Slack** – Residual memory content written to disk
- **Drive Slack** – Residual data from previous files

Table 4: Slack Space vs Unallocated Space

FEATURE	SLACK SPACE	UNALLOCATED SPACE
Location	Inside Allocated Cluster	Outside Allocated Files
Contains Old Data	Yes	Yes
Visible To User	No	No
Forensic Relevance	High	Very High

3.5 TRIM IN SSD

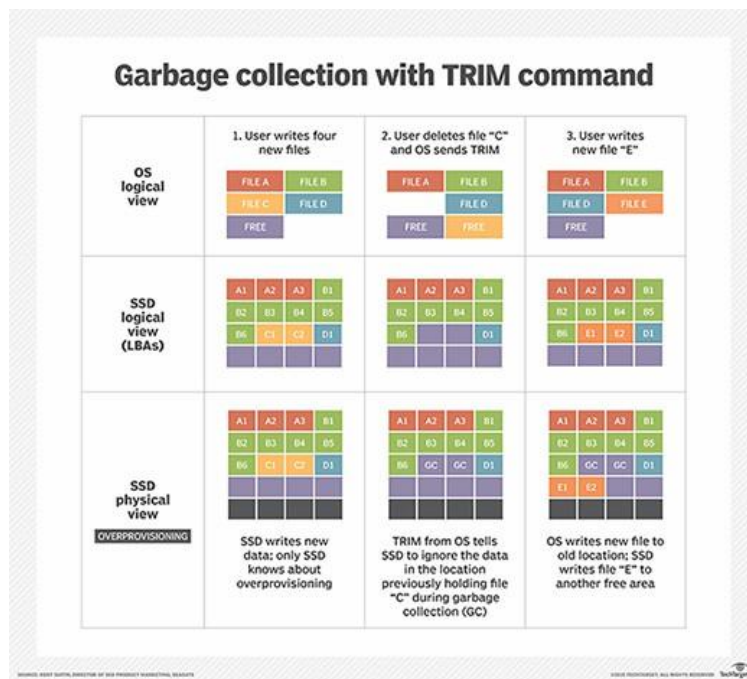


Figure 9: TRIM

TRIM is a command supported by modern SSDs that informs the drive which data blocks are no longer in use and can be erased internally. Unlike HDDs, SSDs actively clear deleted blocks to optimize performance.

Impact of TRIM on Recovery

- Automatically clears deleted blocks
- Reduces data remanence
- Makes forensic recovery difficult

Table 5: HDD vs SSD (Deletion Behaviour)

FEATURE	HDD	SSD WITH TRIM
Logical Deletion	Data Remains	Data Marked For Removal
Physical Erasure	Only On Overwrite	Triggered By Trim
Recovery Probability	High	Low

3.6 OVERWRITING AND RECOVERY PROBABILITY

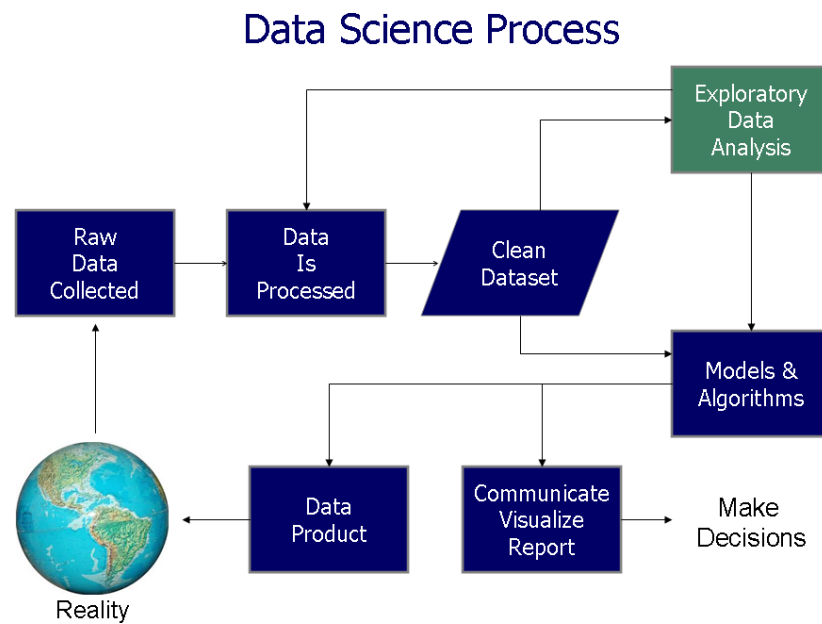


Figure 10: Data Science Process

Overwriting occurs when new data is written over previously stored data blocks. This process directly reduces recovery probability.

Single-pass overwriting replaces data once, while multi-pass overwriting increases destruction reliability.

Table 6: Recovery Probability Based on Overwriting

OVERWRITE LEVEL	RECOVERY PROBABILITY
No Overwrite	Very High
Partial Overwrite	Moderate
Single Pass	Low
Multiple Passes	Extremely Low

CHAPTER 4:

LITERATURE SURVEY

4.1 OVERVIEW OF AUTOPSY

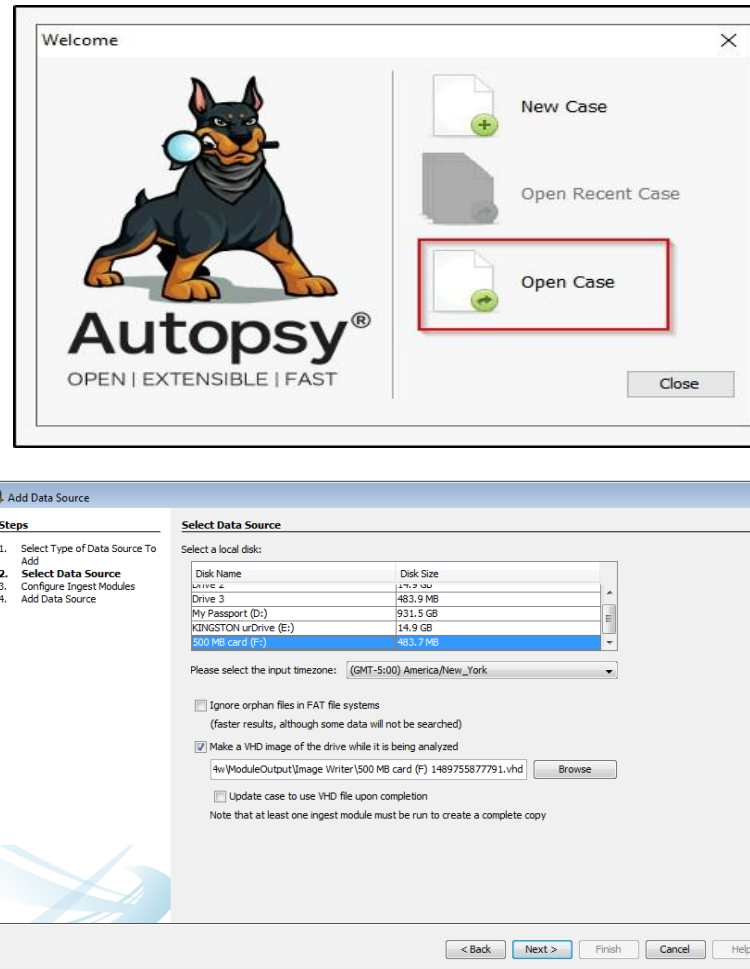


Figure 11: Autopsy

Autopsy is an open-source digital forensics platform widely used by law enforcement agencies, corporate investigators, and cybersecurity professionals for analyzing digital evidence. It provides a user-friendly graphical interface built on top of **The Sleuth Kit**, allowing investigators to conduct forensic examinations without requiring extensive command-line knowledge. The autopsy tool provides support in the analysis of storage media such as hard drives, solid-state drives, USB devices, and various forensic disk image formats including E01, RAW, and AFF. It provides investigators with the ability to recover deleted files, analyze file systems, extract web and internet evidence, conduct

a keyword search, and produce a structured forensic report. These features of the tool enable it to be effective in dealing with various cases of cybercrime.

The autopsy tool stands out as a significant tool in the investigation of crimes through its ability to ensure evidence integrity. It automatically generates cryptographic hash values including MD5, SHA-1, and SHA-256. These hash values are essential in ensuring that evidence does not change during the investigation process. It also provides investigators with logs of all activities during the investigation process.

The autopsy tool would be of significant use in an academic environment or an internship program due to its ability to simplify complex processes in a professional manner. It provides a bridge between theoretical knowledge and investigation processes. It would be essential in the academic environment of those interested in the field of cyber forensics and digital evidence analysis.

4.2 THE SLEUTH KIT (TSK)

```
C:\Program Files\Autopsy-4.11.0\bin>autopsy64.exe --inputPath=R:\work\images\xp-
sp3-v4.E01 --caseName=xpCase --runFromCommandLine=true

C:\Program Files\Autopsy-4.11.0\bin>

The launcher has determined that the parent process has a console and will reuse
it for its own console output.
Closing the console will result in termination of the running program.
Use '-console suppress' to suppress console output.
Use '-console new' to create a separate console window.
Temp Folder for Libraries: C:\Users\User2\AppData\Local\Temp
SleuthkitJNI: loaded libtsk_jni
Embedded jre directory found in: C:\Program Files\Autopsy-4.11.0\jre
Using java binary path: C:\Program Files\Autopsy-4.11.0\jre\bin\java
Processing Autopsy command line options
runFromCommandLine = true
Input file = R:\work\images\xp-sp3-v4.E01
Case name = xpCase
runFromCommandLine = true
Running from command line
Data source path = R:\work\images\xp-sp3-v4.E01
Case name = xpCase
Output directory = C:\Work\Cases
Job processing task finished
```

```
(root@kali)-[/mnt/secret]
# df -h .
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb         9.1M   14K   8.3M   1% /mnt/secret

(root@kali)-[/mnt/secret]
# echo "Hello World" > secret.txt && chmod 666 secret.txt

(root@kali)-[/mnt/secret]
# wget 'https://images.pexels.com/photos/16369267/pexels-photo-16369267.jpe
g?auto-compress&cs=tinysrgb&w=1260&h=750&dpr=1' -o picture.jpeg

(root@kali)-[/mnt/secret]
# ls -la
total 19
drwxr-xr-x 3 root root 1024 May  9 22:33 .
drwxr-xr-x 3 root root 4096 May  9 22:24 ..
drwx----- 2 root root 12288 May  9 22:23 lost+found
-rw-r--r-- 1 root root 442 May  9 22:33 picture.jpeg
-rw-rw-rw- 1 root root 12 May  9 22:28 secret.txt

(root@kali)-[/mnt/secret]
# rm -rf secret.txt picture.jpeg
```

Figure 12: Sleuth Kit

The Sleuth Kit (TSK) is a collection of powerful command-line-based digital forensic analysis tools used to analyze disk images and file systems. Using TSK, digital forensic experts can analyze file systems, recover deleted data, analyze disk partitions, and obtain critical information from digital data. TSK supports various commonly used file systems such as NTFS, FAT32, exFAT, and EXT2/3/4. Using TSK to analyze digital data is only possible if one has sufficient knowledge and experience in executing various commands such as `fls` to list files, `icat` to recover files, and `mmls` to analyze disk partitions. These commands are useful in obtaining accurate digital forensic information from digital data. However, executing these commands is only possible if one has knowledge and experience in using command-line-based digital forensic analysis tools such as TSK.

On the other hand, Autopsy is a graphical user interface used to automate TSK command-line-based digital forensic analysis processes. Using Autopsy to analyze

digital data is possible without executing any command-line-based commands such as `fls` to list files, `icat` to recover files, and `mmls` to analyze disk partitions. These processes are executed in the background and display results in a structured manner. The integration of TSK and Autopsy offers several benefits. TSK provides forensic accuracy and low-level control, while Autopsy enhances usability through automation and visual presentation. Reporting, which must be compiled manually when using TSK alone, is automatically generated in Autopsy with structured formats suitable for legal or academic documentation.

Table 7: Integration Benefits:

FEATURE	TSK (COMMAND LINE)	AUTOPSY (GUI)
Interface	Text-Based	Graphical
Ease Of Use	Requires Expertise	Beginner-Friendly
Automation	Manual Commands	Automated Workflows
Reporting	Manual Compilation	Built-In Report Generation

4.3 ARCHITECTURE OF AUTOPSY

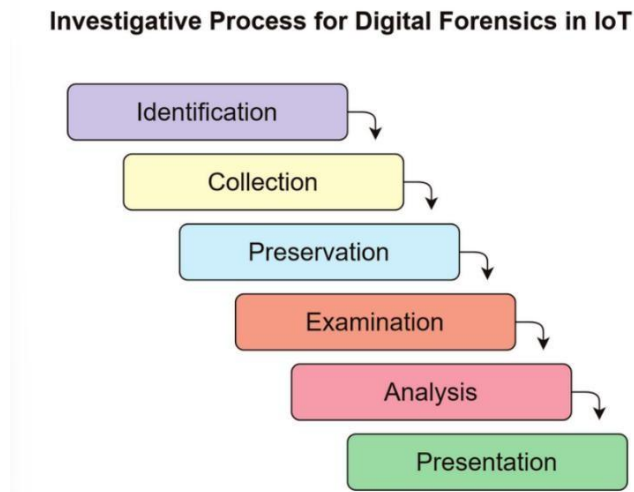
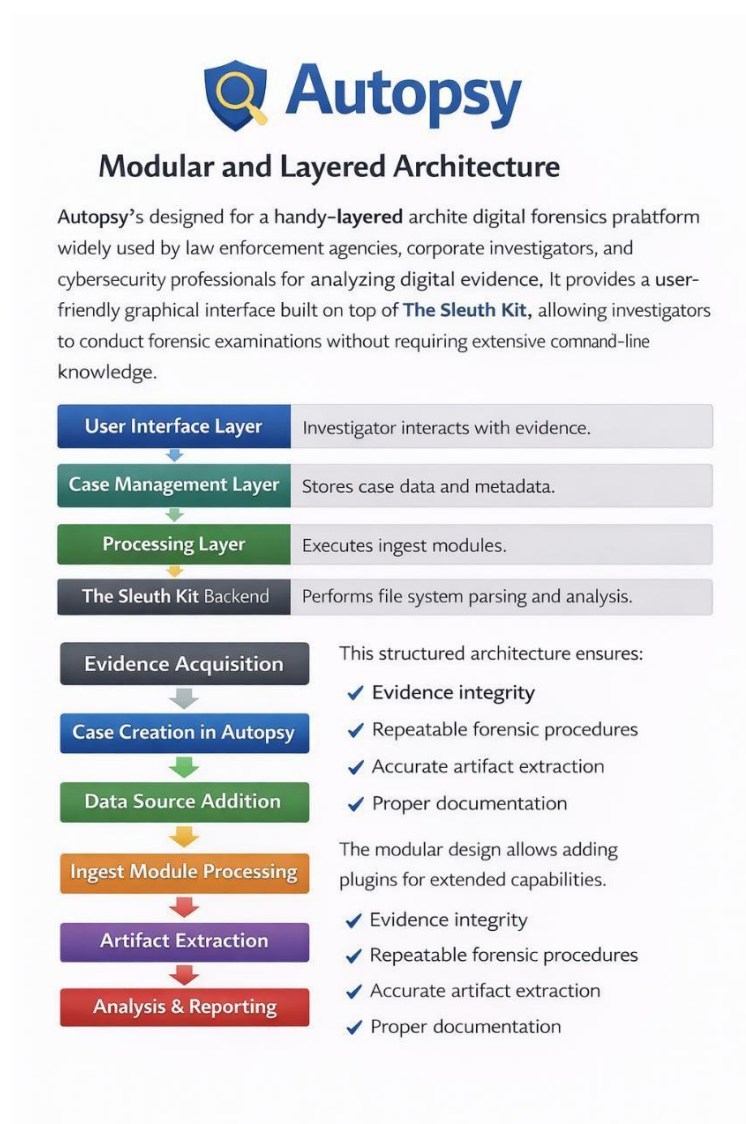


Figure 13: Investigate Process For Digital Forensics

The Autopsy system has a modular and layered structure that is aimed at ensuring scalability, flexibility, as well as a structured approach to digital investigation. The system is composed of four major parts. The User Interface Layer is where the investigator gets to interact with the evidence. The investigator is able to interact with the evidence through a graphical user interface dashboard. The interface is composed of files, data artifacts, timelines, keyword searching, as well as reporting tools.

The second layer is the Case Management Layer. The layer is composed of case data, metadata, as well as investigator notes. The layer is aimed at ensuring that each case is properly isolated as well as documented. The Processing Layer is where ingest modules are executed. The modules are aimed at analyzing data sources as well as identifying relevant digital evidence. The system has The Sleuth Kit Backend as its core. The Sleuth Kit Backend is aimed at performing low-level file system parsing as well as disk analysis. The Sleuth Kit Backend is able to perform technical operations such as reading partitions, recovering deleted files, as well as understanding file system structures. This structured workflow ensures that investigations are methodical and repeatable. It guarantees evidence integrity through hash verification, supports accurate artifact extraction, and maintains proper documentation for legal or academic purposes. Additionally, Autopsy's modular design allows plugins and extensions to be added, enabling customization and enhanced capabilities for specialized forensic requirements.



Picture 14: Architecture

The image describes **Autopsy**, a digital forensics platform with a modular and layered architecture built on The Sleuth Kit, designed to help investigators analyze digital evidence easily through a graphical interface. It consists of multiple layers, including the user interface for interaction, case management for storing data, processing for executing modules, and the Sleuth Kit backend for file system analysis. The workflow involves steps such as evidence acquisition, case creation, adding data sources, processing ingest modules, extracting artifacts, and generating analysis reports. This structured design ensures evidence integrity, repeatable forensic procedures, accurate data extraction, and proper documentation, while also allowing extension through plugins for enhanced capabilities.

4.4 FEATURES AND MODULES OF AUTOPSY

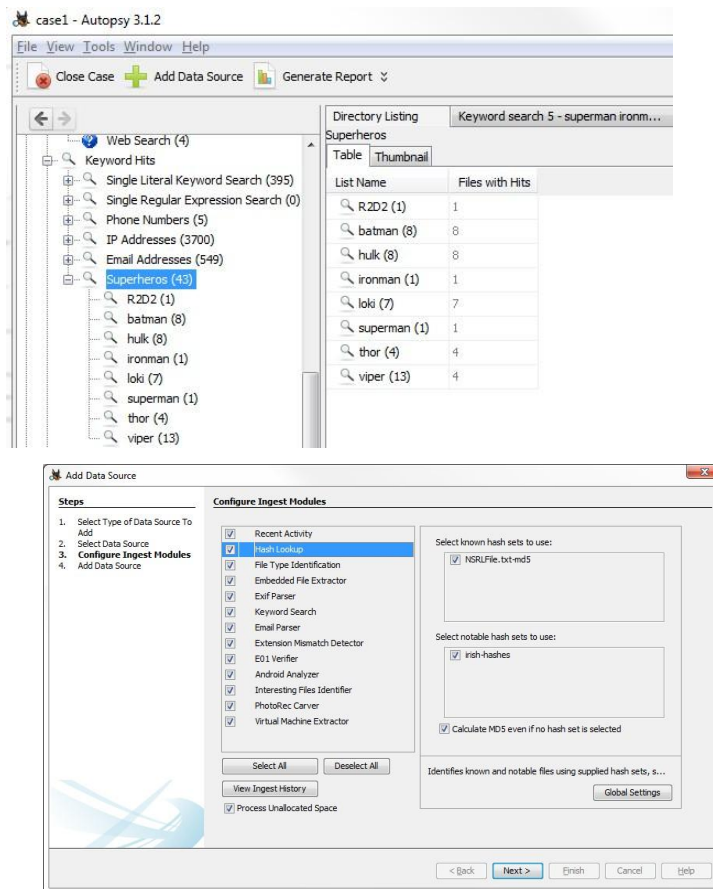


Figure 15: Configure Ingest Modules

Autopsy has various built-in modules that can automate complex forensic activities. These modules are implemented through ingest processes, whereby the system will automatically analyze the added data source to extract relevant information. This minimizes manual work while ensuring accuracy and reliability in digital forensic investigations.

Core Features

One of the features is **Keyword Search**. This allows users to search for suspicious words or phrases on the entire disk image. This is particularly useful in cybercrime investigations where certain words or phrases must be identified on a disk image. For example, certain words or communication content may be used in online communication.

The Web Artifacts Extraction feature allows users to recover information related to web browsing activities such as history, cookies, downloaded files, bookmarks, and cache files. This is particularly useful in cybercrime investigations where users must identify a user's browsing activities. Similarly, the Email Analysis feature allows users to recover email communications and related information. These are particularly useful in fraud or harassment investigations. File Type Identification **uses signature-based analysis rather** than file extensions to determine the true type of a file. This helps detect disguised or renamed files.

Deleted File Recovery

enables the recovery of logically deleted files that still exist in unallocated space. Additionally, **Hash Lookup**

compares file hashes against known databases to identify known good files (such as operating system files) or known malicious files.

Table 8: Module Classification:

MODULE TYPE	PURPOSE
Data Carving	Recovers Files Without Metadata By Scanning Raw Data
File Type Identification	Identifies Files Using Signature Analysis
Hash Lookup	Compares Files Against Known Hash Databases
Recent Activity	Extracts Recent User Activity And System Usage
Android Analyzer	Examines Android Device Backups And Artifacts

Automated Ingest Node (win-4913)

Status: **Running**

Services Status: Case databases up, keyword search up, coordination up, messaging up

Pending Jobs

Case	Data Source	Job Created	Prioritized
gpolon	kw_jr06.img	2016/08/26 15:04:35	☒
alpha	mt02_system.bin	2016/08/26 15:04:35	
alpha	mt02_userdata.bin	2016/08/26 15:04:35	
beta	thunderbird_email_image.dd	2016/08/26 15:04:35	

Prioritize Case

Deprioritize Case

Prioritize Job

Deprioritize Job

Running Jobs

Case	Data Source	Host Name	Stage	Time in Stage
gamma	dump.bin	win-4913	Opening case	9 s

Ingest Progress

Cancel Job

Cancel Module

Completed Jobs

Case	Data Source	Job Created	Job Created	Job Completed	Status
nl	small.img	2016/08/26	2016/08/26 15:04:35	2016/08/26 15:01:46	
alpha	mt01_cache.bin	2016/08/26	2016/08/26 15:04:35	2016/08/26 15:11:07	
beta	green_images.img	2016/08/26	2016/08/26 15:04:35	2016/08/26 15:08:48	
beta	blue_images.img	2016/08/26	2016/08/26 15:04:35	2016/08/26 15:09:56	
alpha	jdk0_jvmchko.bin	2016/08/26	2016/08/26 15:04:35	2016/08/26 15:06:28	

Reprocess Job

Delete Case

Show Case Log

Start

Refresh

Options

Open System Logs Folder

Cluster Metrics

Exit

Add Data Source

Steps

- Select Data Source
- Configure Ingest Modules**
- Add Data Source

Configure Ingest Modules

☒ Recent Activity

☒ Hash Lookup
 ☒ File Type Identification
 ☒ Embedded File Extractor
 ☒ Exif Parser
 ☒ Keyword Search
 ☒ Email Parser
 ☒ Extension Mismatch Detector
 ☒ E01 Verifier
 ☒ Android Analyzer
 ☒ Interesting Files Identifier
 ☒ PhotoRec Carver
 ☒ Virtual Machine Extractor

Select All

Deselect All

View Ingest History

☒ Process Unallocated Space

The selected module has no per-run settings.

Extracts recent user activity, such as Web browsing, recently ...

Global Settings

< Back

Next >

Finish

Cancel

Help

In Autopsy, Ingest Modules are plugins that are automated to process new data sources that are added. In this system, once a disk image or storage device is added to a case, various ingest modules can be selected depending on the needs of the investigation. These plugins will then run in the background while scanning the data source that has been added.

Ingest modules are plugins that are intended to automate various repetitive forensic activities while ensuring accuracy and integrity. Instead of analyzing file systems, databases from a web browser, or metadata manually, various ingest modules are utilized to perform these activities systematically. Common Ingest Modules

- **File Type Identification** – Detects actual file types using signature analysis rather than file extensions.
- **Keyword Search** – Scans the data source for specific words, phrases, or patterns.
- **Hash Lookup** – Compares file hashes against known good or known bad hash databases.
- **Web Artifacts** – Extracts browser history, cookies, downloads, and cached data.
- **Email Parser** – Recovers email messages, attachments, and communication metadata.
- **EXIF Metadata Extractor** – Extracts metadata from images, including timestamps and device information.

During the automated scanning phase, Autopsy utilizes The Sleuth Kit as a backend tool to analyze file systems and extract relevant information. The extracted evidence is grouped together and displayed in an organized manner.

The ingest modules help reduce the workload, speed up the investigation process, and provide consistency in the investigation process. This helps Autopsy become an efficient tool in academic projects, internship case documentation, and professional investigations of cyber crimes.

4.6 TIMELINE ANALYSIS

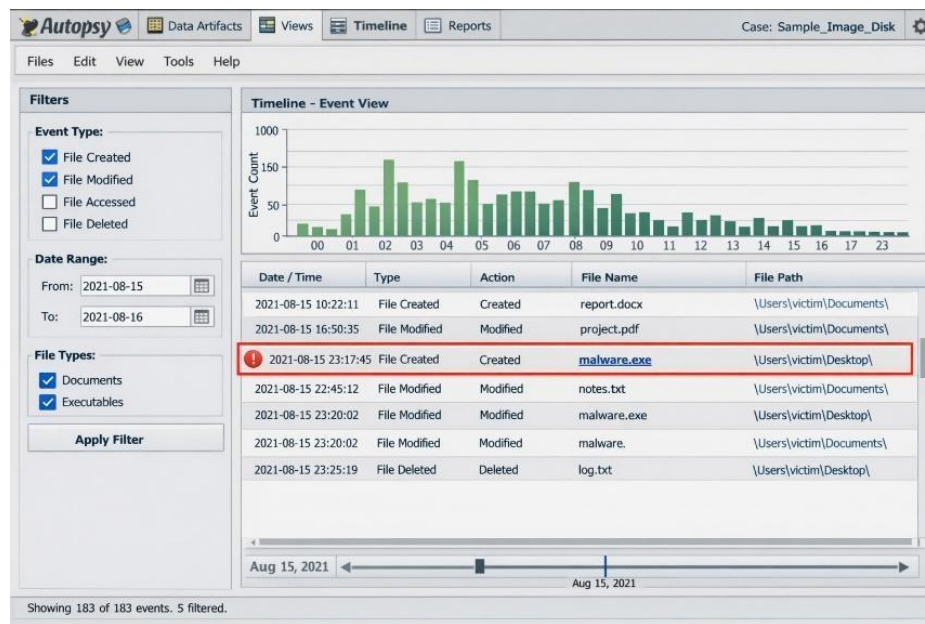


Figure 17: Timeline Analysis

Timeline Analysis is an essential part of a digital forensic investigation since it assists in reconstructing events in chronological order. Autopsy uses timestamp analysis in the following manner: timestamp analysis in Autopsy works by extracting timestamp information from file systems. It helps investigators visualize activities on a particular system or user over a specified time period and helps in recognizing suspicious activity patterns.

Digital devices have the inherent ability to record time stamps whenever a file is created, modified, accessed, or changed. Autopsy collects these time stamps and presents them in a unified format in a single window. It helps investigators analyze multiple events in a single window. There are different types of time stamps used in Autopsy:

- created – It indicates the time of creation of a file.
- modified – It indicates the time of last modification of a file.
- accessed – It indicates the time of last access of a file.
- changed (MFT Record Change) – It indicates the time of last change in the metadata of a file in an NTFS file system.

4.7 REPORTING AND DOCUMENTING

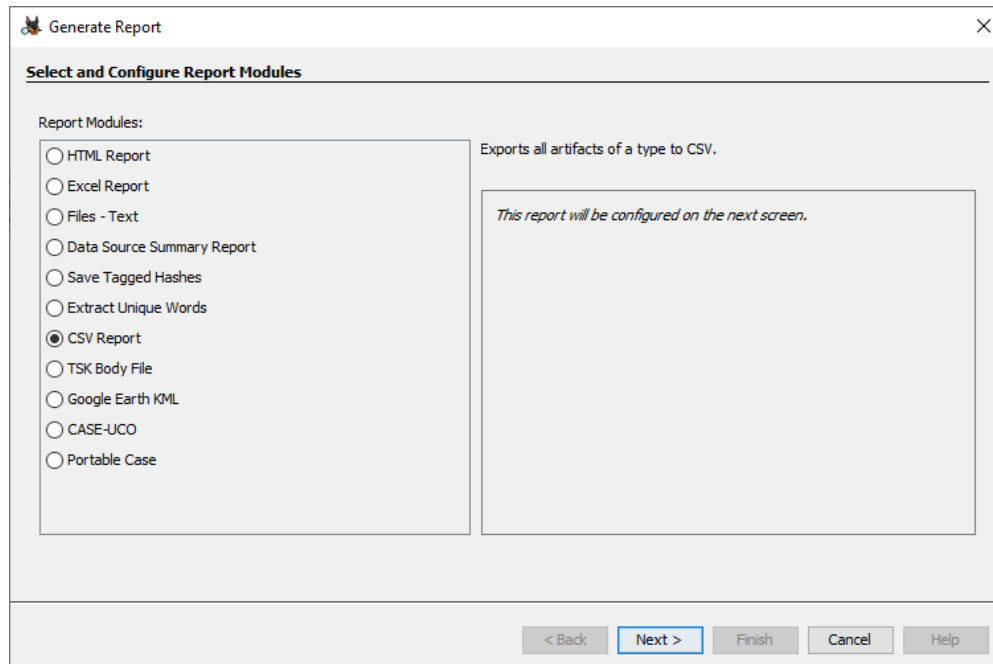


Figure 18: Generate Report – Select And Configure Report Modules

Documentation is the final and most critical phase of any digital forensic investigation. No matter how technically sound an analysis is, it holds little value without proper reporting. In **Autopsy**, reporting and documentation are structured, automated, and court-oriented, ensuring that investigative findings are presented clearly, accurately, and professionally.

After completing the forensic analysis, Autopsy allows investigators to generate reports in multiple formats, including:

- **HTML** – Interactive reports suitable for quick review
- **PDF** – Formal documentation for legal or academic submission
- **CSV** – Structured data for further technical analysis
- **Excel** – Tabular reports for sorting and filtering artifacts

Contents of an Autopsy Report

A standard forensic report generated using Autopsy typically includes:

- **Case Details** – Case number, investigator name, date, and description
- **Evidence Summary** – Information about disk images or data sources examined

- **Extracted Artifacts** – Browser history, emails, deleted files, documents, etc.
- **Hash Values** – MD5/SHA hash verification for evidence integrity
- **Timeline Events** – Chronological reconstruction of activities
- **Investigator Notes** – Observations and conclusions

This structured documentation ensures clarity and maintains the chain of custody, which is essential in cybercrime investigations.

Proper documentation ensures:

- **Legal admissibility** – Reports must meet evidentiary standards in court.
- **Transparency** – Clear explanation of tools, methods, and findings.
- **Professional presentation** – Structured and readable format.
- **Academic evaluation compliance** – Meets institutional project requirements.

CHAPTER 5:

DESIGN

5.1 UNIFIED MODELLING LANGUAGE

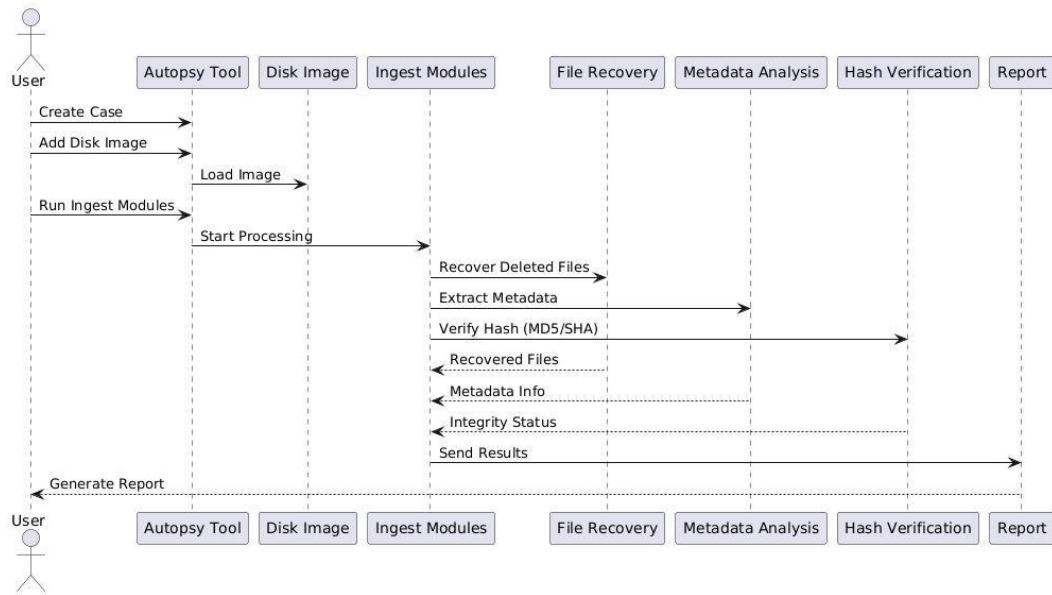


Figure 19: UML

This diagram illustrates the digital forensic investigation workflow using Autopsy, where a user creates a case, adds and loads a disk image, and runs ingest modules to start processing; the system then performs file recovery, extracts metadata, and verifies file integrity using hash values (MD5/SHA), after which the recovered files, metadata information, and integrity status are compiled and sent to generate the final forensic report.

5.2 CLASS DESIGN

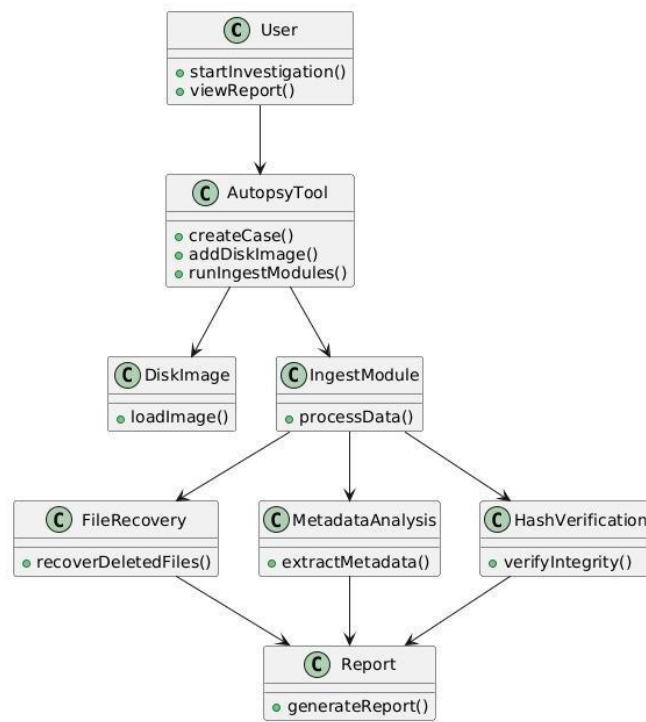


Figure 20: Class Design

This diagram represents a Digital Forensics System where a user interacts with various functions used in investigating digital evidence. The system allows the user to create a case and add a disk image as the primary data source, after which different analysis processes can be performed. These include verifying hash integrity to ensure data authenticity, analysing metadata to extract hidden details like timestamps and file origins, and recovering deleted files for evidence retrieval. The user can also run ingest modules to process and organize data efficiently, and finally generate reports to document findings. Overall, the system helps in collecting, analysing, and presenting digital evidence in a structured and reliable way, making it useful for forensic investigations.

5.3 USE CASE

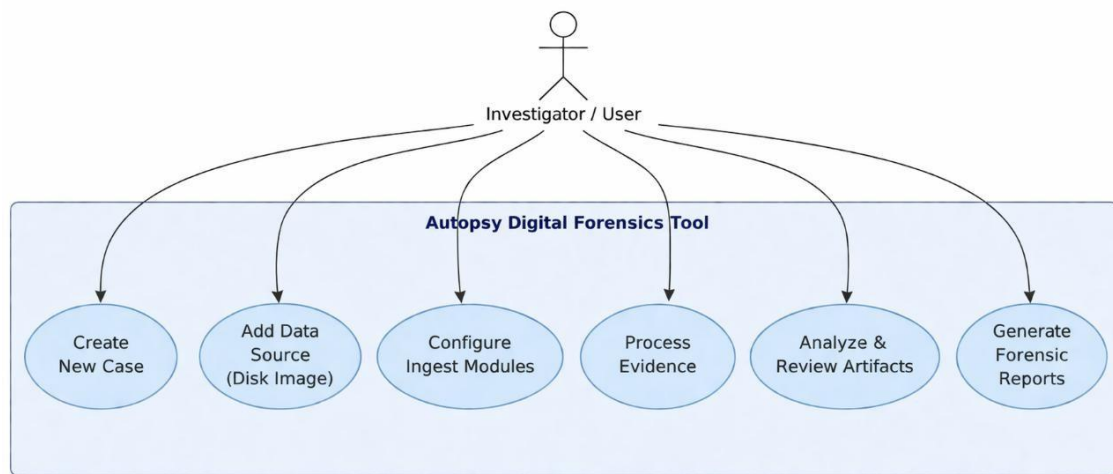


Figure 21: Use Case

This class diagram represents the structure of a digital forensic system using Autopsy, where the User initiates the investigation and views reports, interacting with the Autopsy Tool that manages core functions like creating a case, adding a disk image, and running ingest modules; the Disk Image class handles loading of evidence data, while the Ingest Module processes the data and coordinates with File Recovery to recover deleted files, Metadata Analysis to extract relevant metadata, and Hash Verification to ensure data integrity, with all results ultimately passed to the Report class to generate the final forensic report.

5.4 SEQUENCE DIAGRAM

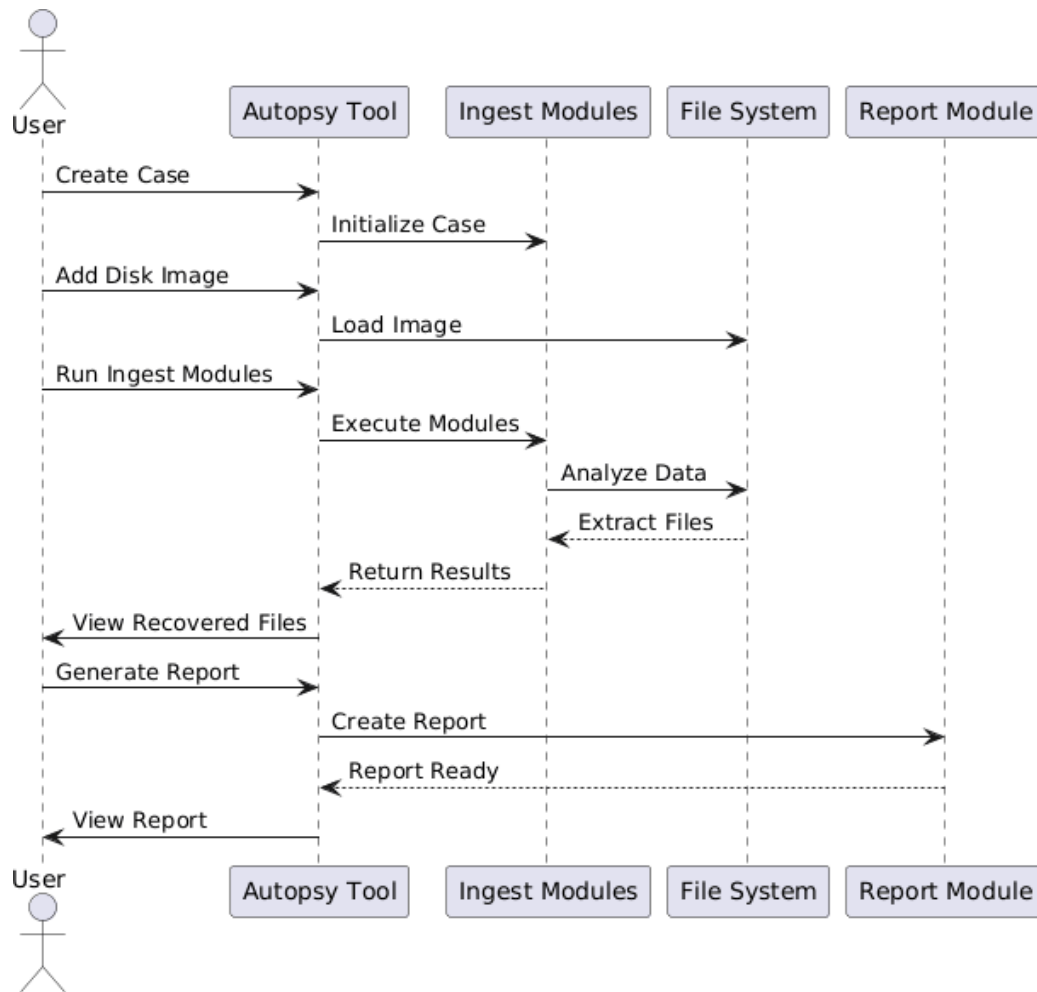


Figure 23: Sequence Diagram

The diagram illustrates the workflow of a digital forensic investigation using the Autopsy tool. It begins with the user creating a case and adding a disk image, after which the Autopsy tool initializes the case and loads the image into the file system. The user then runs ingest modules, which execute various analysis processes to examine the data and extract relevant files. The results of this analysis are returned to the Autopsy tool, allowing the user to view recovered files. Finally, the user generates a report, which is created by the report module and marked as ready for viewing. This sequence shows how data flows between the user, Autopsy tool, ingest modules, file system, and report module to complete a forensic investigation.

CHAPTER 6:

IMPLEMENTATION

Project Demonstration Video:

https://drive.google.com/drive/folders/1sCQukuk4_hr1EfLiBr7jLFJ74wI2-tFy?usp=sharing

6.1 SOFTWARE INSTALLATION

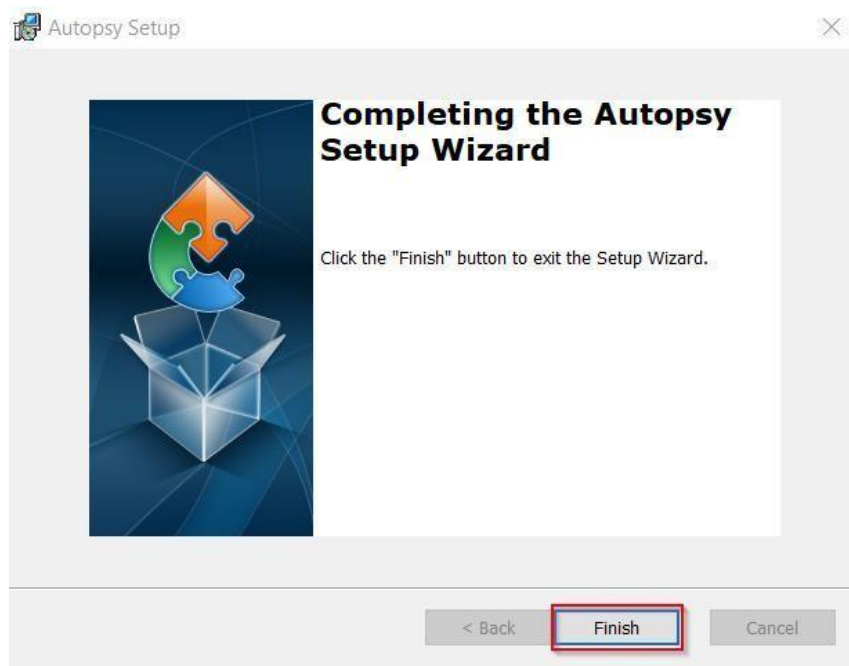


Figure 24: Software Installation

The primary forensic tool used in this internship project is **Autopsy**, which integrates with The Sleuth Kit.

Installation Steps:

1. Download Autopsy (64-bit version).
2. Install Java (if not pre-installed).

3. Run the installer with administrative privileges.
4. Complete setup wizard.
5. Launch Autopsy and verify functionality.

Additional Tools Installed:

- FTK Imager (for disk imaging)
- Hash verification utilities
- Win Hex (optional hex-level inspection)

Table 9: Installation Verification Checklist:

STEP	STATUS
Java Installed	Yes
Autopsy Installed	Yes
Case Creation Test	Yes
Ingest Module Test	Yes

Proper installation ensures smooth forensic workflow and accurate artifact extraction.

6.2 DISK IMAGING PROCESS

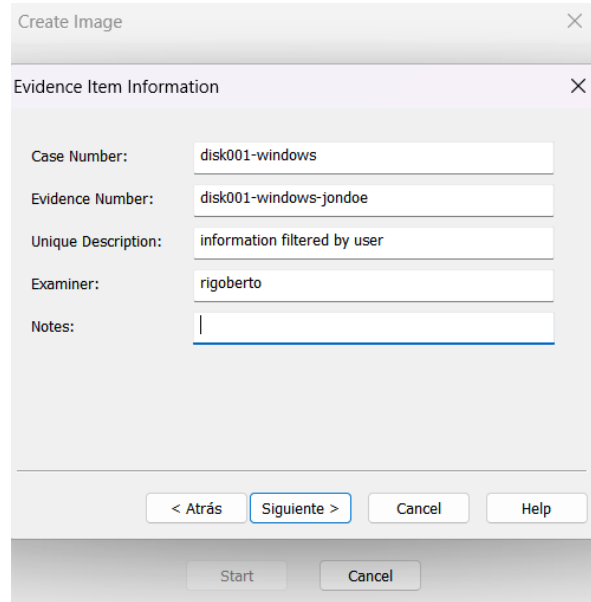


Figure 25: Creating Image

Disk imaging is the process of creating a bit-by-bit copy of a storage device. It preserves original evidence and prevents alteration.

Imaging Tools Used:

- FTK Imager
- Write Blocker (if available)

Image Formats Comparison:

Table 10: Disk Image

FORMAT	DESCRIPTION	ADVANTAGE
RAW	Exact Bit Copy	Simple And Universal
E01	Encase Format	Compression + Metadata
AFF	Advanced Forensic Format	Efficient Storage

Hash values generated during imaging ensure integrity. If hash before and after imaging matches, evidence remains untampered.

6.3 CREATION OF TEST DATA

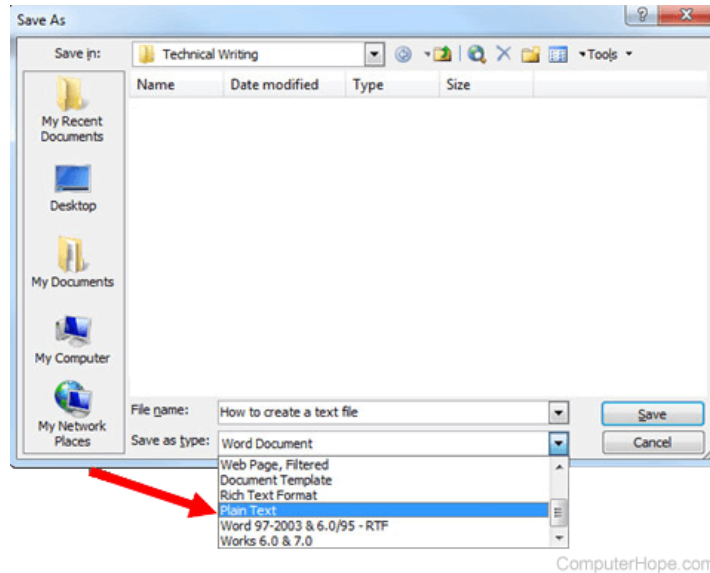


Figure 26: Save As Dialog Box – Selecting Plain Text Format

To simulate a realistic forensic scenario, controlled test data was created.

Types of Data Created:

- Text documents (.txt, .docx)
- Images (.jpg, .png)
- PDF files
- ZIP archives
- Browser history entries
- Downloaded sample files

Sample Scenario:

- Create a folder named "Confidential Data"
- Add documents and images
- Insert USB device and copy files
- Access files multiple times

Table 11: Purpose of Test Data:

ACTIVITY	FORENSIC OBJECTIVE
File Creation	Observe Metadata Generation
File Modification	Track Timestamp Changes
Usb Transfer	Analyze Removable Media Artifacts
Internet Browsing	Extract Web Artifacts

6.4 DELETION SCENARIOS

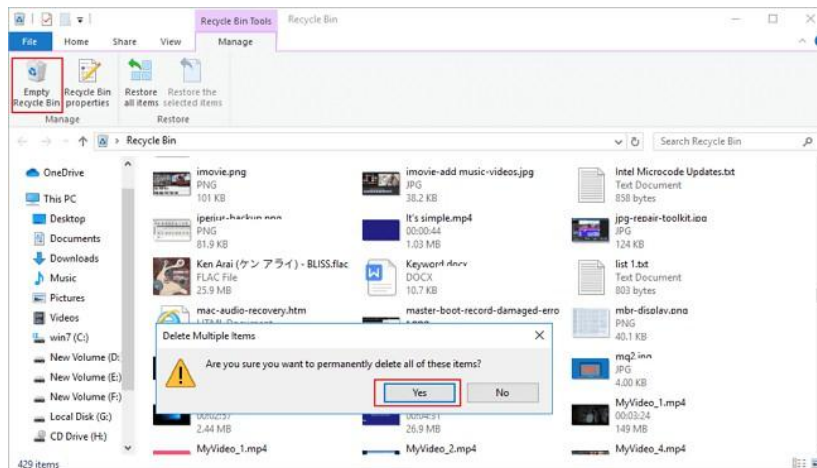


Figure 27: Recycle Bin – Permanent Delete Confirmation Window

Different deletion techniques were tested to analyze recovery possibilities.

Scenario 1: Normal Delete

File → Deleted → Moves to Recycle Bin → Recoverable

Scenario 2: Shift + Delete

File → Bypasses Recycle Bin → Marked as Unallocated → Recoverable (Until Overwritten)

Scenario 3: Formatting USB

Drive → Quick Format → Metadata Removed → Partial Recovery Possible

Scenario 4: SSD with TRIM

File Deleted → TRIM Command Sent → Blocks Cleared → Low Recovery Probability

Testing these scenarios provided practical insight into file system behavior and digital evidence recovery techniques using Autopsy.

Table 12: Comparison Table:

DELETION METHOD	RECOVERY CHANCE	FORENSIC DIFFICULTY
Recycle Bin	Very High	Low
Shift Delete	High (If Not Overwritten)	Medium
Quick Format	Moderate	High
Ssd Trim	Very Low	Very High

Deletion Flowchart:

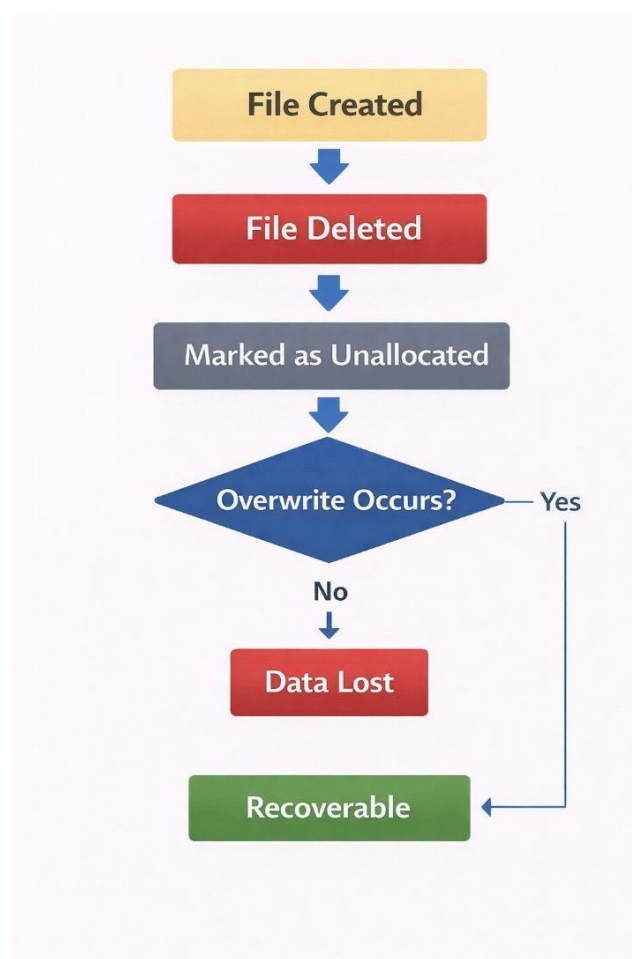


Figure 28: Deletion Flowchart

CHAPTER 7:

EXPERIMENTAL PROCEDURE

7.1 CASE CREATION IN AUTOPSY

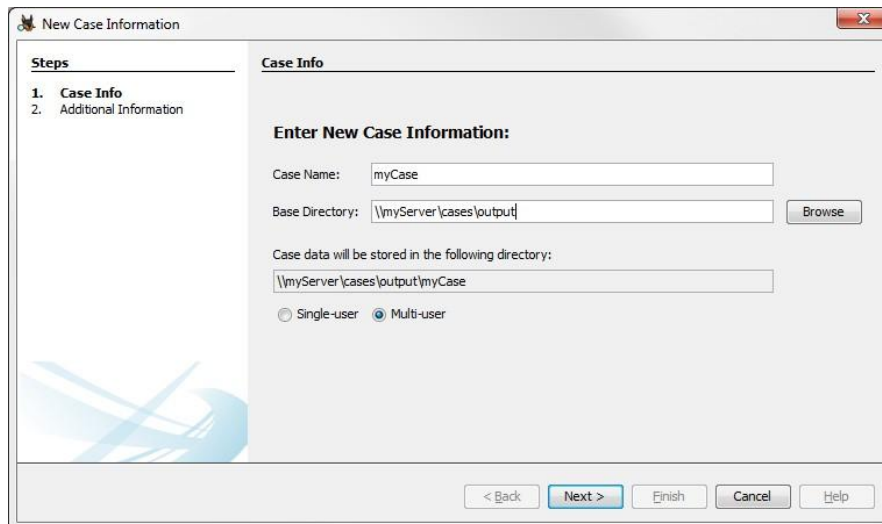


Figure 27: New Case Information

The first step in the experimental procedure is creating a new forensic case in Autopsy. Case creation organizes all evidence, analysis results, logs, and reports under a structured directory.

Steps Followed:

1. Launch Autopsy.
2. Click **Create New Case**.
3. Enter Case Name (e.g., *Deleted_File_Recovery_Test*).
4. Specify Case Directory.
5. Enter Examiner Details.
6. Finish Setup.

Creating a structured case ensures proper documentation and maintains professional forensic standards. It also supports chain of custody principles

Table 13: Importance of Case Creation:

COMPONENT	PURPOSE
Case Name	Identifies Investigation
Case Directory	Stores Evidence & Reports
Examiner Details	Maintains Accountability
Logs	Records Investigative Actions

7.2 ADDING DISK IMAGE

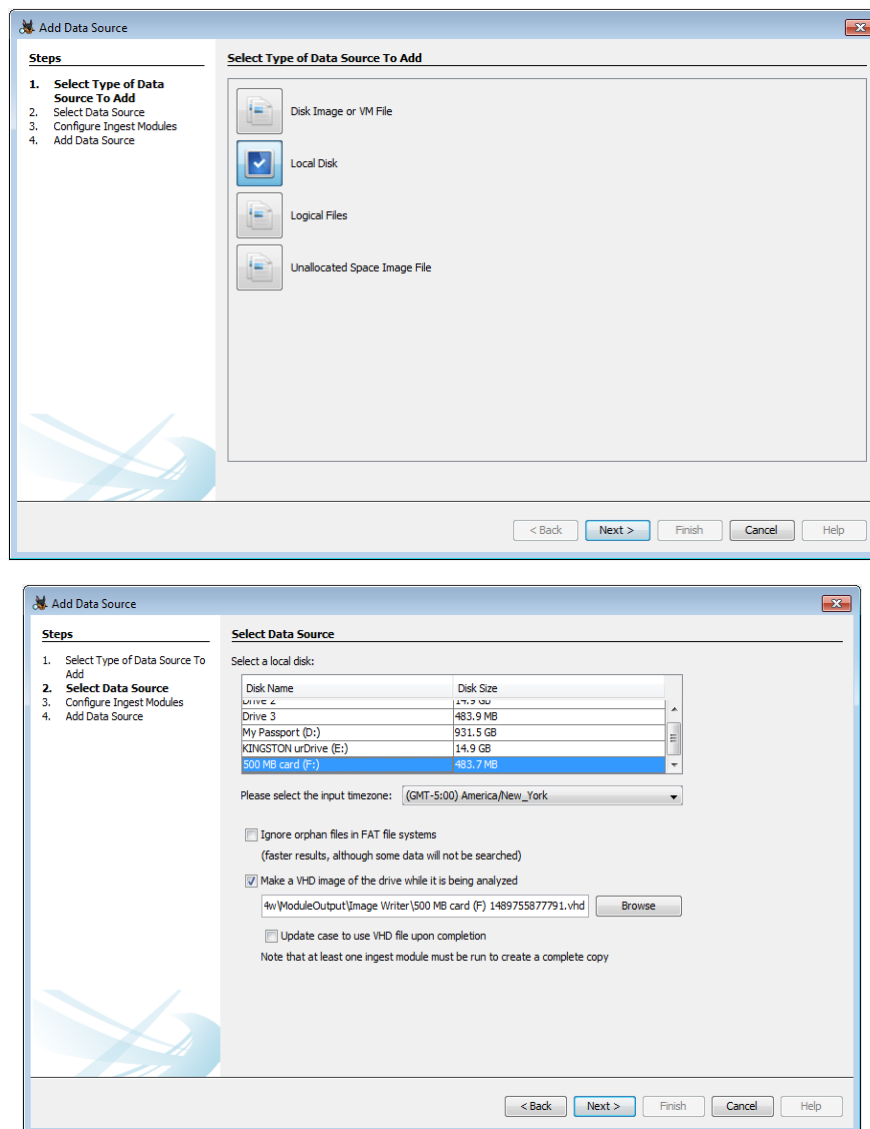


Figure 29: Adding Disk Image

After case creation, the next step is adding the forensic disk image.

Procedure:

1. Select **Add Data Source**.
2. Choose **Disk Image or VM File**.
3. Browse and select the previously created image (E01/RAW).
4. Confirm hash verification.
5. Proceed to ingest module selection.

Why Disk Images Are Used:

- Prevents alteration of original evidence
- Enables repeated analysis
- Preserves metadata and deleted data

Table 14: Verification Step:

CHECK	PURPOSE
Hash Match	Confirms Integrity
File System Detection	Identifies Ntfs/Fat
Partition Layout	Detects Hidden Partitions

7.3 RUNNING INGEST MODULES

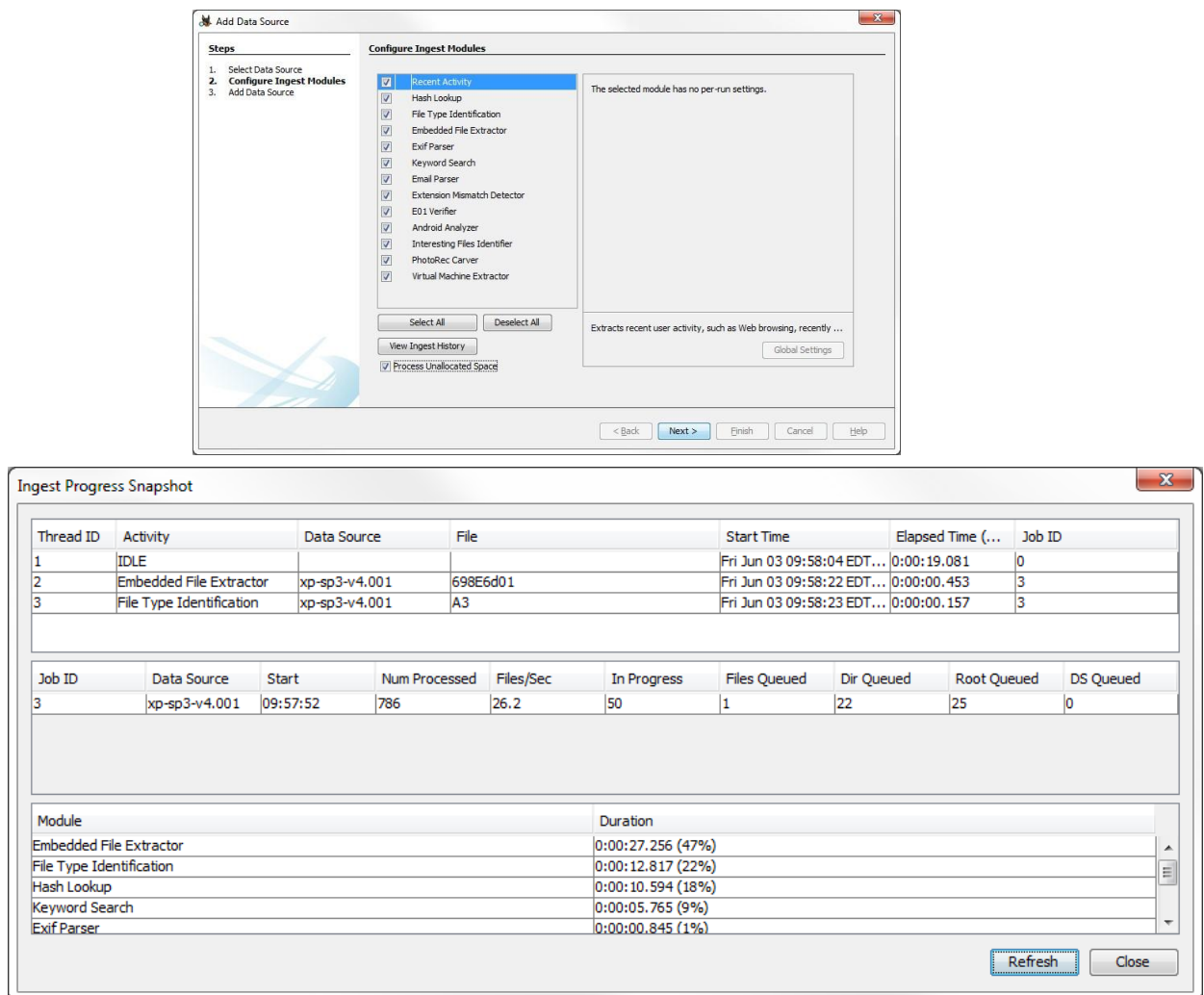


Figure 30: Ingest Progress Snapshot

Ingest modules automate artifact extraction and significantly reduce manual workload.

Selected Ingest Modules:

- File Type Identification
- Keyword Search
- Hash Lookup
- Recent Activity
- Web Artifacts
- Data Carving

Workflow:

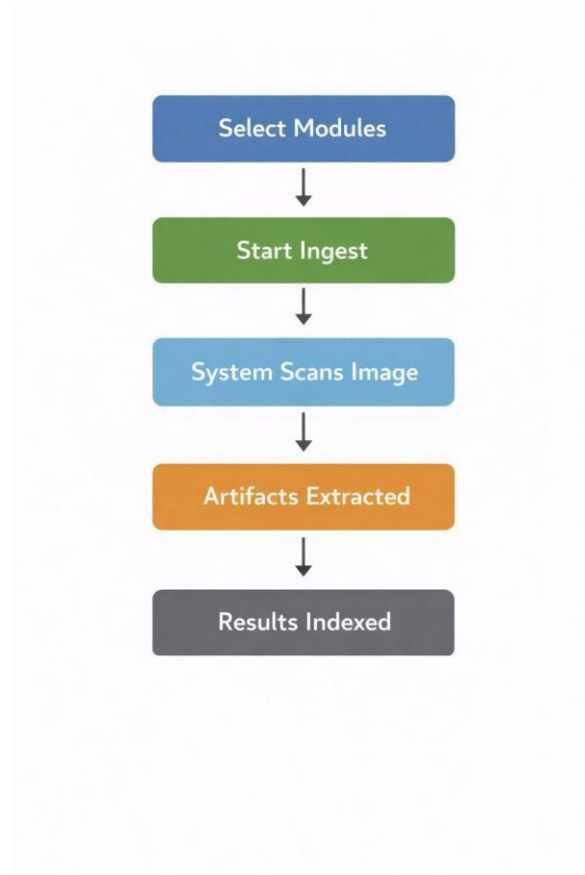


Figure 31: Running Ingest modules Flowchart

Table 15: Benefits of Ingest

ADVANTAGE	EXPLANATION
Automation	Saves Time
Accuracy	Reduces Human Error
Artifact Categorization	Organized Evidence
Scalability	Handles Large Datasets

The ingest process may take time depending on image size and system configuration.

[illegible]

Methods Used:

- 67

Deletion Behaviour in NTFS:

Observations:

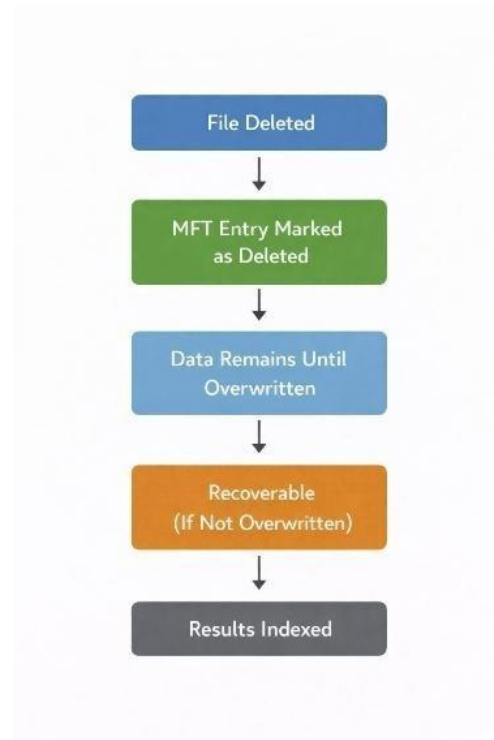


Figure 33: Identifying Deleted Files Flowchart

Table 16: NTFS

DELETION TYPE	RECOVERY STATUS
Normal Delete	Fully Recoverable
Shift Delete	Recoverable
Quick Format	Partially Recoverable
Ssd With Trim	Rarely Recoverable

Autopsy successfully recovered test documents deleted using normal and Shift+Delete methods.

7.5 FILE EXTRACTION

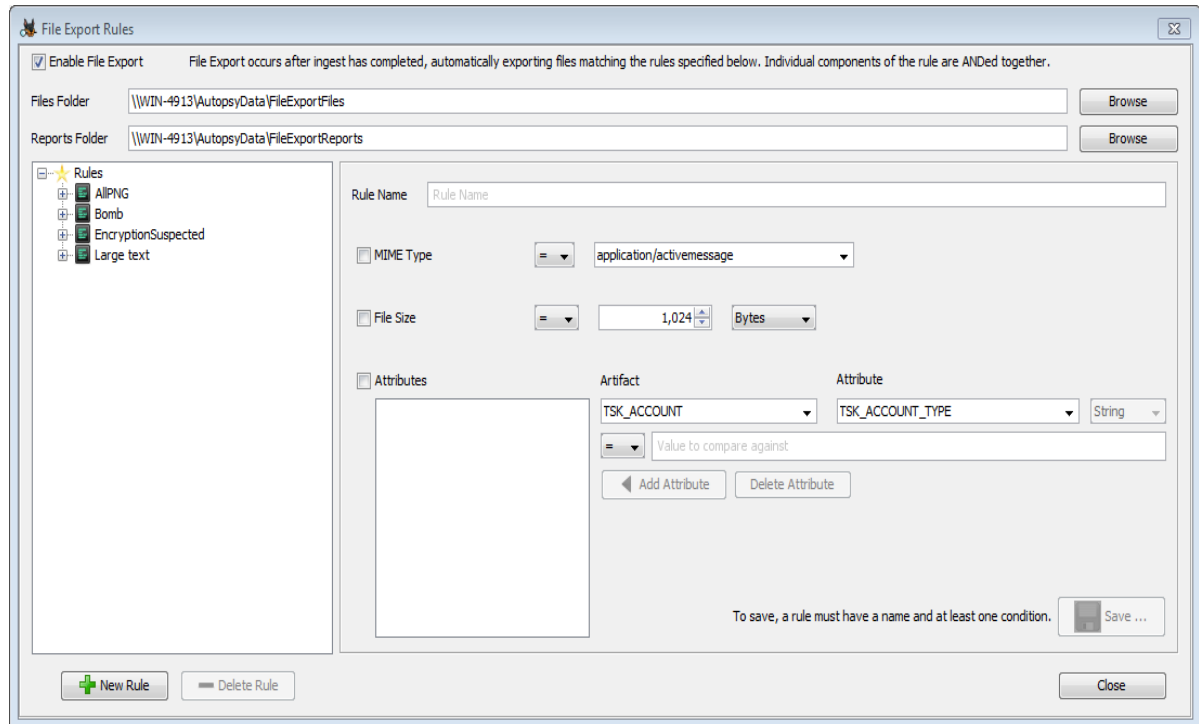


Figure 34: File Expert Rules

After identifying deleted or suspicious files, they were extracted for further examination.

Extraction Steps:

1. Right-click on file.
2. Select **Extract File(s)**.
3. Choose destination folder.
4. Save extracted copy.

Table 17: Purpose of Extraction:

PURPOSE	EXPLANATION
Evidence Preservation	Keeps Original Image Intact
Detailed Analysis	Open In External Tools
Documentation	Attach To Forensic Report

Extracted files were opened in a secure environment to verify content and confirm successful recovery.

CHAPTER 8:

RECOVERY RESULTS

8.1 RECOVERY RATE BY FILE TYPE

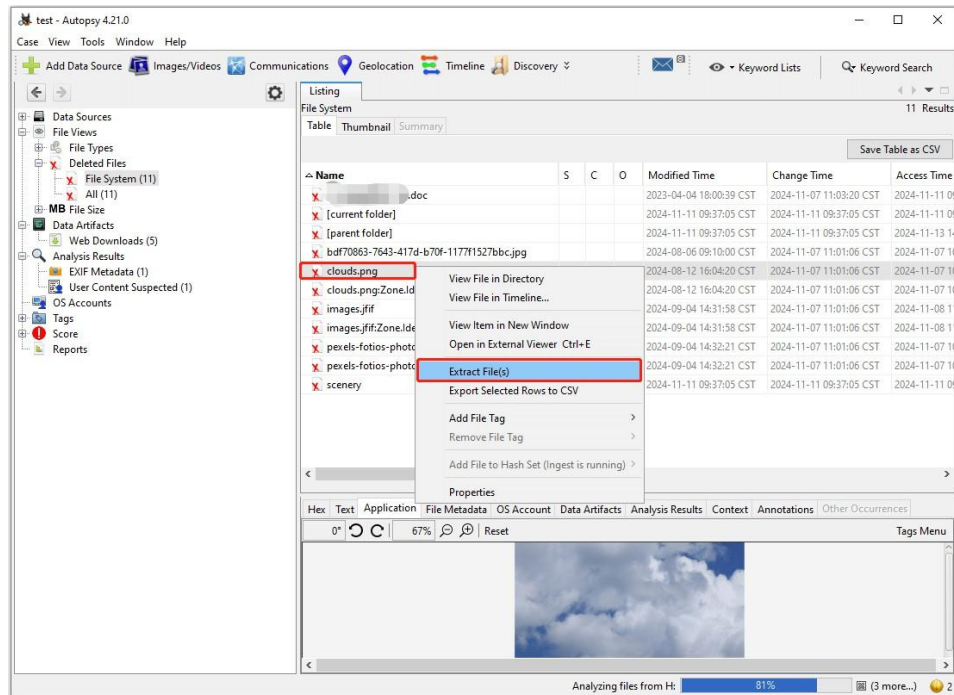


Figure 35: Extract File Option

The recovery experiment evaluated how different file types respond to deletion and recovery attempts. Results showed that recovery success varies depending on file size, storage behavior, and overwrite conditions.

Observations:

- **Text files (.txt, .docx)** – High recovery rate
- **Images (.jpg, .png)** – Moderate to high recovery
- **PDF files** – High recovery if not fragmented
- **ZIP archives** – Lower recovery if partially overwritten
- **Large media files** – Lower recovery due to fragmentation

Table 18: Recovery Rate Table:

FILE TYPE	RECOVERY RATE	REASON
Text Documents	95%	Small Size, Less Fragmentation
Images	85%	Stored Contiguously
Pdfs	90%	Structured Format
Zip Files	70%	Sensitive To Corruption
Large Videos	50–60%	Highly Fragmented

Smaller and less fragmented files demonstrated higher recovery success during the experiment

Flow Insight:

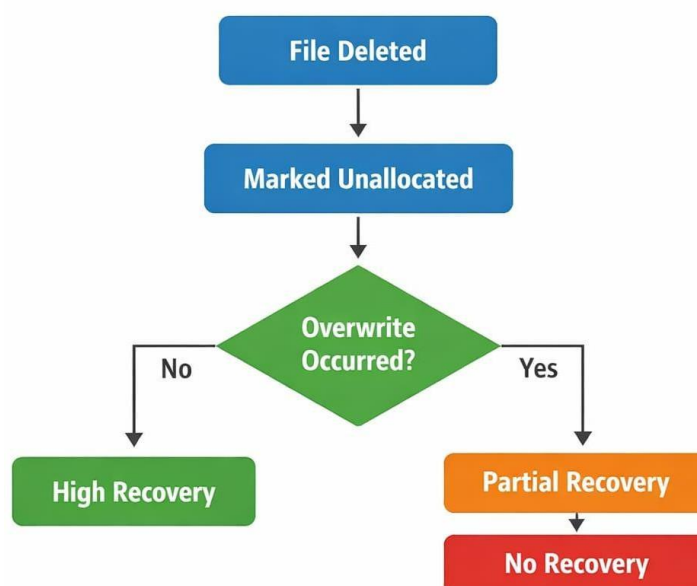


Figure 36: Recovery Data Of Filetype Flowchart

8.2 INTEGRITY ANALYSIS

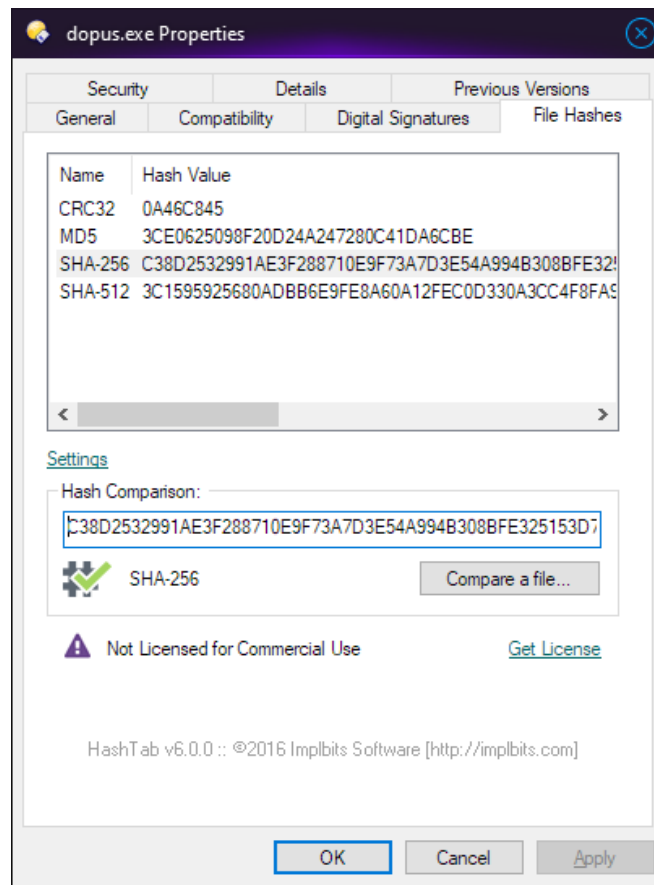


Figure 37: Integrity Analysis

Integrity analysis verifies that recovered files remain unchanged from their original state. Hash Comparison Procedure:

1. Generate hash during disk imaging.
2. Generate hash after file extraction.
3. Compare both values.

Interpretation:

- Matching hash values confirm file authenticity.
- Mismatch indicates partial overwrite or corruption.
- No hash = unrecoverable data.

Integrity analysis confirmed that most small test files were recovered without modification.

Table 19: Integrity Verification Table:

FILE	ORIGINAL HASH	RECOVERED HASH	STATUS
Doc1.Txt	Match	Match	Verified
Image1.Jpg	Match	Match	Verified
Report.Pdf	Match	Match	Verified
Archive.Zip	Partial	Mismatch	Corrupted

Drive C:	\$MFT after	\$MFT before	
Offset	0	1	2 3 4 5 6 7 8 9 A B C D E F
00321400	46 49	4C 45	30 00 03 00 44 6C 18 4C 04 00 00 00
00321410	39 02 01 00	38 00	01 00 C8 01 18 4C 00 04 00 00
00321420	00 00 00 00	00 00 00 00	07 00 00 00 85 0C 00 00
00321430	0B 00 00 00	00 00 00 00	10 00 00 00 60 00 00 00
00321440	00 00 00 00	00 00 00 00	48 00 00 00 18 00 00 00
00321450	DE 3F D5 5B BF A1 CB 01	DC F0 7B BC 6D A1 CB 01	b?0[0iE U\$0kmiE
00321460	06 7C 01 55 81 A6 CB 01	AA 01 B1 4E 81 A6 CB 01	U E a ±N E
00321470	20 20 00 00	00 00 00 00	00 00 00 00 00 00 00 00
00321480	00 00 00 00	5D 01 00 00	00 00 00 00 00 00 00 00
00321490	08 FC 52 58 00 00 00 00	30 00 00 00 78 00 00 00	uRX 0 x
003214A0	00 00 00 00	00 00 06 00	5A 00 00 00 18 00 01 00
003214B0	EF 25 00 00	00 00 01 00	DE 3F D5 5B BF A1 CB 01
003214C0	DC F0 7B BC 6D A1 CB 01	DC F0 7B BC 6D A1 CB 01	U\$0kmiE U\$0kmiE
003214D0	7B EE 4E 41 81 A6 CB 01	00 40 00 00 00 00 00 00	xiNA E @
003214E0	9C 34 00 00	00 00 00 00	20 20 00 00 00 00 00 00
003214F0	0C 03 61 00 61 00 72 00	64 00 76 00 61 00 72 00	14 a a r d v a r
00321500	6B 00 2E 00 74 00 78 00	74 00 00 00 00 00 00 00	k t x t
00321510	80 00 00 00	48 00 00 00	01 00 00 00 00 04 00
00321520	00 00 00 00	00 00 00 00	03 00 00 00 00 00 00 00
00321530	40 00 00 00	00 00 00 00	00 40 00 00 00 00 00 00
00321540	9C 34 00 00	00 00 00 00	9C 34 00 00 00 00 00 00
00321550	41 04 B4 7D B9 00	00 00 00 00 68 00 00 00	A } ^ 1 h
00321560	01 0F 40 00 00 00 05 00	00 00 00 00 00 00 00 00	@
00321570	00 00 00 00	00 00 00 00	60 00 00 00 00 00 00 00
00321580	00 10 00 00	00 00 00 00	2E 00 00 00 00 00 00 00
00321590	2E 00 00 00	00 00 00 00	5A 00 6F 00 6E 00 65 00
003215A0	2E 00 49 00	64 00 65 00	6E 00 74 00 69 00 66 00
003215B0	69 00 65 00	72 00 00 00	41 01 B8 7D B9 00 00 00 00
003215C0	FF FF FF FF	82 79 47 11	9C 34 00 00 00 00 00 00

Timestamp analysis helps reconstruct user activity patterns.

- Created
- Modified
- Accessed
- MFT Changed

EVENT	OBSERVATION
Creation Time	Preserved After Deletion
Modified Time	Indicates Content Editing
Access Time	Shows User Interaction
Deletion Marker	Reflected In Mft

CHAPTER 9:

CHALLENGES AND LIMITATIONS

9.1 RISKS OF IMPROPER DELETION

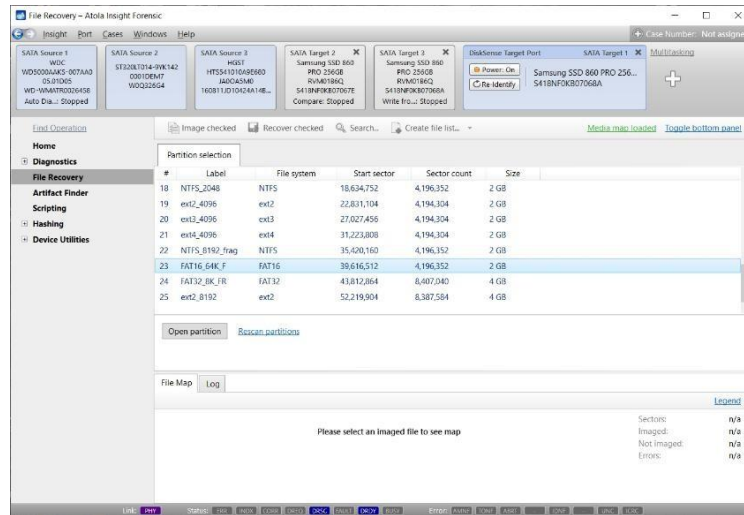


Figure 39: File Recovery

Improper deletion refers to the practice of removing files using standard operating system commands (Delete or Shift+Delete) without secure wiping. During the internship experiment, it was observed that such deletion methods only remove file references from the file system, while the actual data remains in storage blocks until overwritten.

Security Risks:

- **Data Remanence:** Residual data persists after deletion.
- **Unauthorized Recovery:** Attackers can recover confidential documents.
- **Identity Theft:** Personal information may be extracted.
- **Corporate Espionage:** Business strategies and trade secrets may be exposed.
- **Legal Consequences:** Failure to securely dispose of data can violate compliance regulations.

9.2 INSIDER THREAT PERSPECTIVE

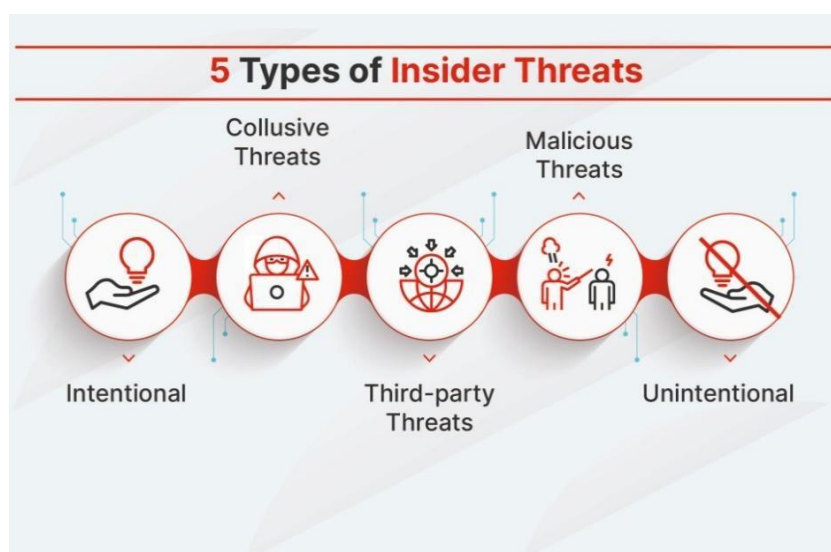


Figure 40: Insider Threats

From an insider threat perspective, improper deletion can be exploited by employees, contractors, or privileged users who have legitimate system access.

Table 21: Insider Risk Categories:

INSIDER TYPE	SECURITY IMPACT
Malicious Insider	Intentionally Recovers Deleted Confidential Files
Negligent Insider	Accidentally Exposes Recoverable Data
Compromised Insider	Credentials Misused To Access Deleted Data

Since insiders already have system-level access, they can:

- Retrieve deleted documents.
- Access unallocated space.
- Extract hidden artifacts.
- Modify logs to conceal activity.

Mitigation Strategies:

- Strict access control policies
- Activity logging and monitoring
- Least privilege enforcement
- Secure data wiping procedures

The experiment demonstrated that even deleted internal documents could be recovered if not securely erased, highlighting serious insider security risks.

9.3 ETHICAL HACKING IMPLICATIONS

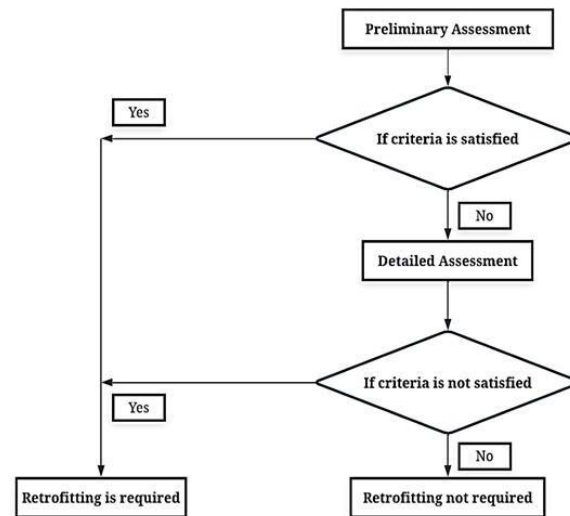


Figure 41: Flowchart

Ethical hackers simulate real-world attacks to identify vulnerabilities before malicious actors exploit them. Improper data deletion is a critical vulnerability often discovered during security assessments.

During Security Testing:

Implications:

- Organizations may unknowingly expose sensitive data.
- Poor disposal policies can fail compliance audits.
- Security maturity level may be questioned.

Ethical hacking assessments often include:

- Recovery attempts from formatted drives
- Analysis of unallocated space
- Verification of secure wiping procedures

This highlights the importance of integrating secure deletion policies into cybersecurity frameworks.

9.4 CORPORATE SECURITY POLICIES

CYBER SECURITY POLICY

This Cyber Security Policy includes guidelines and provisions for security measures to help mitigate cyber security risk. It applies to all company employees, contractors, volunteers, and anyone who has permanent or temporary access to the company's systems and hardware.

1. CONFIDENTIAL DATA

Confidential data is valuable and is to be kept secret. Company confidential data includes:

- Unpublished financial information
- Data of customers/partners/vendors
- Patents, formulas or new technologies
- Customer lists (existing and prospective)

All employees are obliged to protect this data.

2. PROTECT PERSONAL AND COMPANY DEVICES

When employees use their digital devices to access company emails or accounts, they introduce security risk to company data. Employees are to keep both their personal and company-issued computer, tablet and cell phone secure. To keep these devices secure:

- Keep all devices password protected.
- Choose and upgrade a complete antivirus software.
- Do not leave devices exposed or unattended.
- Install security updates of browsers and systems monthly or as soon as updates are available.
- Log into company accounts and systems through secure and private networks only.

Employees are advised to avoid accessing internal systems and accounts from other people's devices or lending their own devices to others.

When new hires receive company-issued equipment, they will receive instructions for:

- Disk encryption setup
- Password management tool setup
- Installation of antivirus/anti-malware software

Employees are to follow instructions to protect their devices and refer to company Security Specialists/Network Engineers with any questions.

Organizations must implement formal data destruction and retention policies to prevent recoverable data exposure.

Table 22: Essential Policy Components:

POLICY ELEMENT	PURPOSE
Data Classification	Identify Sensitivity Level
Retention Period	Define Data Lifecycle
Secure Disposal Procedure	Specify Wiping Standards
Audit & Compliance	Ensure Policy Enforcement
Incident Reporting	Manage Exposure Events

Recommended Controls:

- Mandatory multi-pass wiping for HDDs
- Encryption-based deletion for SSDs
- Cryptographic erase for enterprise storage
- Documented destruction certificates

Corporate policies should align with regulatory frameworks such as data protection laws and industry compliance standards.

9.5 CLOUD DATA EXPOSURE RISKS

Home > Create a resource > Create a virtual machine >

Create a new disk ...

Create a new disk to store applications and data on your VM. Disk pricing varies based on factors including disk size, storage type, and number of transactions. [Learn more](#) ⓘ

Name *	deletetest_DataDisk_0
Source type * ⓘ	None (empty disk) ▼
Size * ⓘ	1024 GiB Premium SSD LRS Change size
Encryption type *	(Default) Encryption at-rest with a platform-managed key ▼
Enable shared disk	☐ Yes ☒ No
Delete disk with VM	☐

Picture 4: New Disk Creation Cloud

Cloud computing introduces additional complexities in secure deletion and data exposure.

Key Cloud Risks:

- **Multi-Tenancy:** Data from multiple users stored on shared infrastructure.
- **Virtual Disk Reuse:** Improper sanitization may expose previous tenant data.
- **Backup Persistence:** Deleted files may remain in backup systems.
- **Cross-Border Data Storage:** Legal jurisdiction complications.

Cloud Deletion Flow:

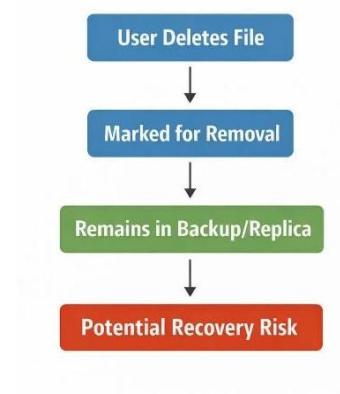


Figure 43: Cloud Data Exposure Flowchart

Risk Mitigation Measures:

- Encryption-at-rest
- Cryptographic erase mechanisms
- Strong Service Level Agreements (SLAs)
- Cloud provider compliance verification

Cloud environments require trust in service providers' deletion processes. Therefore, encryption-based deletion is often the most reliable security mechanism in cloud infrastructures.

CHAPTER 10:

TEST CASES

The test cases were developed based on real-world scenarios of data loss and cyber forensic investigations. Each test case includes:

1. Test Case ID
2. Objective
3. Input/Scenario
4. Expected Result
5. Actual Result

TEST CASES AND OBSERVATIONS

Test Case 1: Normal File Deletion (Recycle Bin)

Test Case ID: TC_01

Objective: To verify recovery of files deleted using standard deletion

Input: File deleted using Delete key

Process: File moved to Recycle Bin and analyzed

Expected Result: File should be fully recoverable

Actual Result: File was successfully recovered

Observation:

Files deleted normally remain intact in storage and can be easily recovered unless permanently removed.

Test Case 2: Permanent Deletion (Shift + Delete)

Test Case ID: TC_02

Objective: To analyze recovery after permanent deletion

Input: File deleted using Shift + Delete

Process: File marked as unallocated space

Expected Result: Recoverable until overwritten

Actual Result: File recovered successfully

Status: Pass

Observation:

Even after permanent deletion, file data remains in unallocated space, making recovery possible.

Test Case 3: Quick Format

Test Case ID: TC_03

Objective: To evaluate recovery after formatting

Input: Drive formatted using quick format

Process: Metadata removed, data analyzed using Autopsy

Expected Result: Partial recovery

Actual Result: Some files recovered, some corrupted

Status: Partial Pass

Observation:

Quick formatting removes file references but not actual data, enabling partial recovery.

Test Case 4: SSD with TRIM Enabled

Test Case ID: TC_04

Objective: To test recovery limitations in SSD storage

Input: File deleted from SSD

Process: TRIM command clears blocks

Expected Result: Very low recovery

Actual Result: Files were not recoverable

Status: Pass

Observation:

TRIM significantly reduces data remanence, making forensic recovery difficult.

Test Case 5: Overwriting Scenario

Test Case ID: TC_05

Objective: To analyze the impact of overwriting

Input: Deleted file overwritten with new data

Process: Storage blocks reused

Expected Result: Reduced recovery

Actual Result: Files partially or fully unrecoverable

Status: Partial Pass

Observation:

Overwriting directly affects recovery probability and can permanently destroy data.

CHAPTER 11

CONCLUSION

11.1 SUMMARY OF FINDINGS

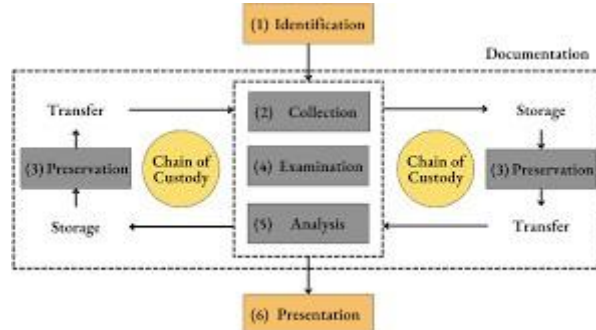


Figure 44: Findings

This internship project focused on **deleted file recovery, storage behavior analysis, and forensic investigation using Autopsy**. Through theoretical study and practical experimentation, several important findings were observed:

Major Findings:

1. Standard Deletion Is Not Secure

Files deleted using normal methods (Delete / Shift+Delete) remain recoverable until overwritten.

2. File Recovery Depends on Storage Type

- HDD → High recovery probability
- SSD (TRIM Enabled) → Very low recovery probability

3. File Type Influences Recovery Success

Smaller files showed higher recovery rates compared to fragmented or large media files.

4. Hash Verification Confirms Integrity

MD5 and SHA256 hashing successfully validated the authenticity of recovered files.

5. Metadata Remains Valuable After Deletion

NTFS timestamps enabled reconstruction of user activity timelines.

11.2 LEARNING OUTCOMES



Figure 45: Learning Outcomes

This internship provided both technical and analytical learning experiences aligned with BCA cybersecurity specialization.

Technical Skills Gained:

- Case creation and management in Autopsy
- Disk imaging procedures
- Running ingest modules
- Deleted file identification
- File extraction and analysis
- Metadata and timestamp examination
- Hash-based integrity verification

Analytical Skills Developed:

- Logical reasoning in forensic investigations
- Evidence documentation practices
- Understanding storage technologies (HDD vs SSD)
- Evaluating recovery limitations

Professional Skills:

- Report preparation
- Structured documentation
- Understanding legal and ethical compliance
- Applying cybersecurity principles practically

This internship strengthened foundational knowledge in digital forensics and enhanced readiness for advanced studies such as MSc in Cyber Forensics.

11.3 INDUSTRY RELEVANCE

Digital forensics plays a critical role in modern cybersecurity operations.

Table 23: Industry Applications:

INDUSTRY SECTOR	FORENSIC IMPORTANCE
Corporate Enterprises	Insider Threat Investigations
Banking & Finance	Fraud Detection
Healthcare	Data Breach Response
Law Enforcement	Cybercrime Investigations
It Companies	Security Audits

With increasing cybercrimes such as ransomware, insider data theft, and corporate breaches, forensic expertise is highly demanded.

Organizations rely on forensic professionals to:

- Investigate data leaks
- Recover deleted evidence
- Validate digital integrity
- Support legal proceedings

The skills developed in this internship directly align with industry requirements in cybersecurity and digital investigation domains.

11.4 REFERENCES

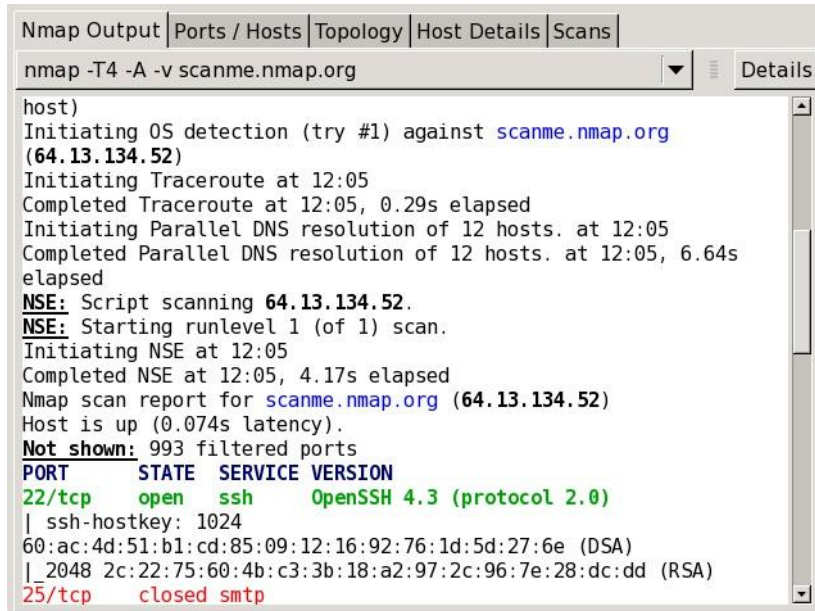
The following references were used in preparing this internship report:

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11.5 APPENDICES

Appendices A –

Tool Commands Used



```

Nmap Output | Ports / Hosts | Topology | Host Details | Scans
nmap -T4 -A -v scanme.nmap.org
host)
Initiating OS detection (try #1) against scanme.nmap.org (64.13.134.52)
Initiating Traceroute at 12:05
Completed Traceroute at 12:05, 0.29s elapsed
Initiating Parallel DNS resolution of 12 hosts. at 12:05
Completed Parallel DNS resolution of 12 hosts. at 12:05, 6.64s elapsed
NSE: Script scanning 64.13.134.52.
NSE: Starting runlevel 1 (of 1) scan.
Initiating NSE at 12:05
Completed NSE at 12:05, 4.17s elapsed
Nmap scan report for scanme.nmap.org (64.13.134.52)
Host is up (0.074s latency).
Not shown: 993 filtered ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 4.3 (protocol 2.0)
| ssh-hostkey: 1024
60:ac:4d:51:b1:cd:85:09:12:16:92:76:1d:5d:27:6e (DSA)
|_ 2048 2c:22:75:60:4b:c3:3b:18:a2:97:2c:96:7e:28:dc:dd (RSA)
25/tcp    closed smtp
  
```

Figure 46: Nmap Service Version Detection Scan Output

Appendices B –

Sample Case Screenshots

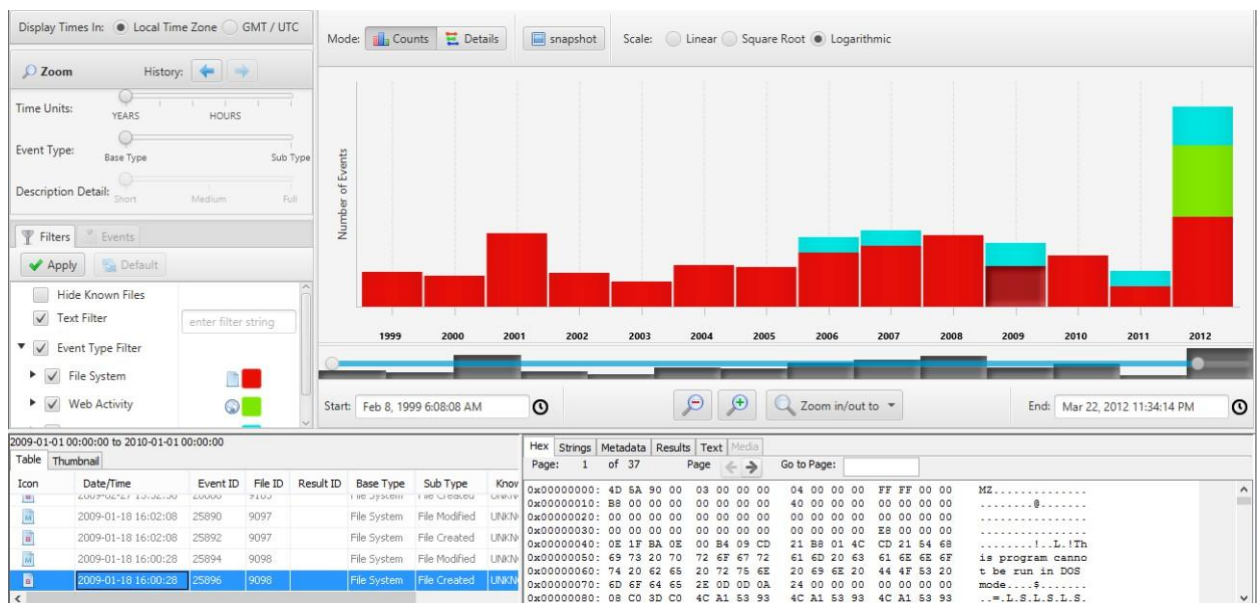


Figure 47: Autopsy Timeline Analysis Of Recovered Artifacts

Appendices C –

Extracted Artifacts

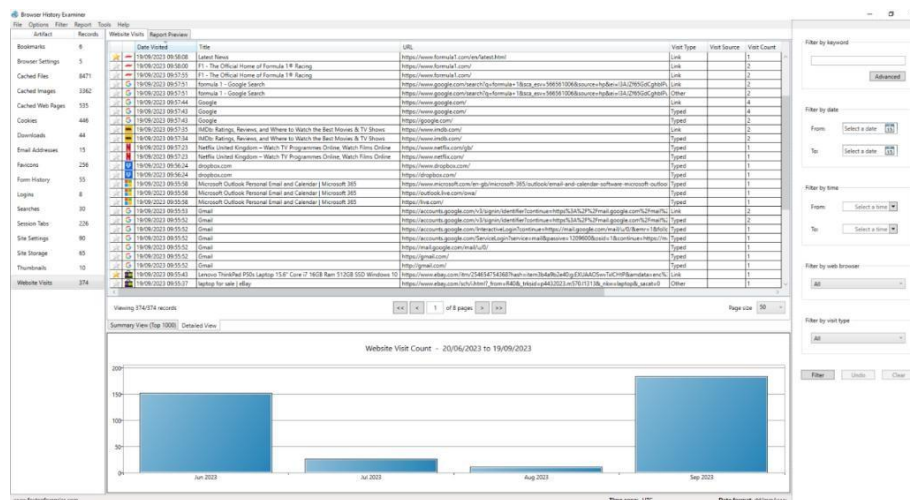


Figure 48: Browser History Artifacts Extracted From Disk Image

Appendices D –

Data Recovery Experiment Results

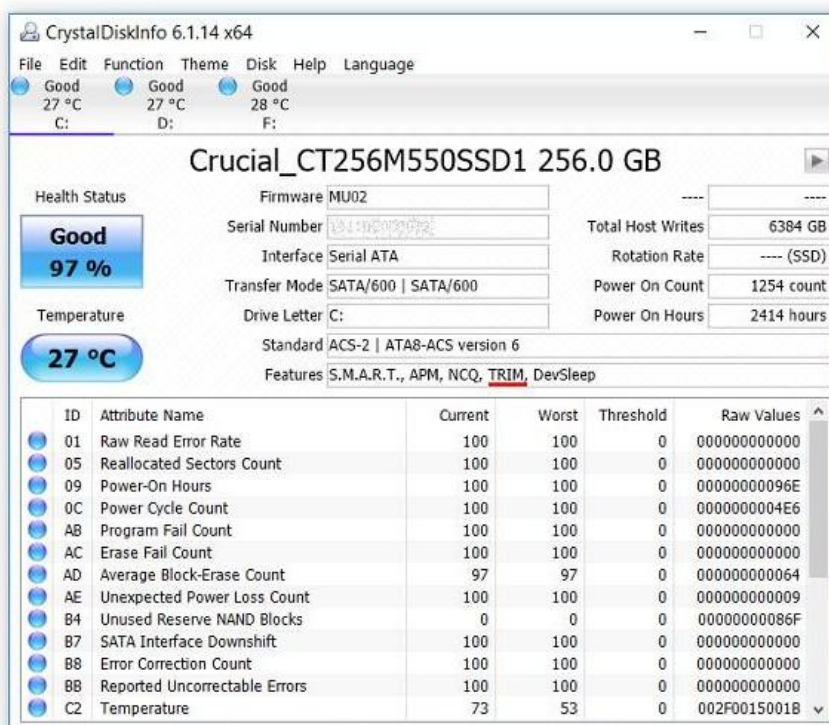


Figure 49: TRIM Impact On SSD Data Recovery